LLMs Can Play (Global) Games

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Abstract

The communication channel that enables coordination is also the channel through which regimes suppress it. I embed seven large language models in the Morris–Shin (2003) regime change global game, conveying private signals as natural-language intelligence briefings. Across 1,800 country–periods, join rates track the theoretical attack mass (mean $r = +0.80$, $p < 0.001$ for every model). The sigmoid survives replacing the signal-to-text mapping with tanh, linear, or step alternatives ($r \approx 0.83$); an eight-condition narrative stakes ladder shifts joining monotonically from 21% to 79% (57.6 pp). I exploit this regularity as a platform for information design: censorship and public-signal manipulations shift coordination sharply, while the ambiguity design points in the predicted direction but is imprecisely estimated. The central finding concerns surveillance: monitored communications shift expressed joining by -9.7 pp across five models while the elicited second-order mean shifts by less than 1 pp. A degraded-messages control—generic text replacing LLM messages, no surveillance framing—drops joining by 21 pp, confirming the mechanism is information degradation, not deterrence. Agents self-censor, peer messages become less informative and less action-oriented, and coordination collapses even though the scalar second-order elicitation barely moves.

JEL: C72, C92, D82, D83, P16
Keywords: global games, regime change, LLM agents, information design, Bayesian persuasion, belief-action wedge, preference falsification

1 Introduction

The information channel is a trap. In the canonical global game of regime change (Morris and Shin, 2003), private signals alone resolve equilibrium multiplicity (Carlsson and van Damme, 1993; Frankel et al., 2003)—but real coordination crises, from bank runs (Diamond and Dybvig, 1983) to currency attacks (Obstfeld, 1996) to political upheaval (Angeletos et al., 2007), also feature communication among agents. Any channel that transmits information about others' willingness to act is also the channel through which regimes suppress it. This paper augments the Morris–Shin framework with an explicit pre-play communication network and documents how surveillance exploits that channel.

Global game models are highly tractable theoretically but notoriously difficult to test. Laboratory experiments can manipulate signal precision and disclosure rules, but only in highly stylized numeric environments that abstract away the rich, multi-source information characterizing real coordination problems (Heinemann et al., 2004, 2009; Szkup and Trevino, 2020). Field data from actual crises confounds strategic behavior with institutional differences and unobserved information structures. The LLM platform manipulates the same information-structure levers—precision, correlation, and disclosure—within a richer natural-language setting at scale.

I address this gap by embedding seven large language models (LLMs) spanning six architecture families in the Morris and Shin (2003) regime change game. Each agent receives a private signal $x_i = \theta + \varepsilon_i$, translated into a natural-language intelligence briefing whose content shifts monotonically with signal strength across eight evidence domains. No explicit payoff table is provided—the stakes are embedded in the narrative, forcing agents to form beliefs from language. The design enables what neither lab experiments nor field data can: testing information design predictions under near-full model complexity, at scale.

Three contributions. First, across 1,800 country–periods and seven models, LLM agents implement threshold policies closely consistent with the unique Bayesian Nash equilibrium. Join rates track the theoretical attack mass with mean $r = +0.80$ ($p < 0.001$ for every model). The alignment is robust to the generator's functional form: replacing the logistic direction slider with tanh, linear, or step functions preserves $r \approx 0.83$, ruling out the hypothesis that the sigmoid is mechanically produced by the mapping architecture. Domain ablation further separates channels: removing coordination-language domains slightly raises mean joining (+2.8 pp) while leaving the sigmoid essentially unchanged, whereas removing regime-state domains lowers joining by 3.4 pp and visibly attenuates the signal-response curve. The evidentiary contrast is therefore in

the preserved versus weakened threshold relation, not in the 0.6 pp gap between the mean shifts alone. An eight-condition narrative cost ladder—holding briefings fixed and varying only a prepended cost/benefit header from "existential consequences" to "zero personal cost"—shifts mean joining monotonically from 21.3% to 78.9% (57.6 pp), with paraphrase variants tracking their originals and a placebo header falling below baseline. The shifted-grid B/C sweep then serves mainly as a centering check, while the fixed-briefing ladder provides the cleaner evidence that strategic-stakes framing in otherwise fixed text moves behavior.

Second, this behavioral regularity converts the experiment into a testbed for information design (Kamenica and Gentzkow, 2011; Goldstein and Huang, 2016). Manipulating signal structure—precision, correlation, or disclosure rules—shifts coordination outcomes as Bayesian persuasion theory predicts. Public signal injection and upper/lower censorship produce large directional shifts; the stability design points in the predicted direction but is statistically ambiguous under the pre-specified H5 test. A publicness $\times$ authority factorial separates the strategic complementarity channel (agents overweight common-knowledge signals) from an authority-cue channel (agents defer to official sources): authority amplifies joining under public framing (+5.4 pp) but not under private framing (-1.7 pp), consistent with strategic complementarity driving the milder header-only publicness effect.

Third, and most importantly, I show that the communication channel that enables coordination is exploitable by an authoritarian planner. Surveillance produces a *belief–action wedge*: expressed joining shifts by -9.7 pp across five models while the elicited second-order mean shifts by less than 1 pp. A degraded-messages control—replacing LLM-generated messages with generic uninformative text, without any surveillance framing—drops joining by 21 pp, confirming that the mechanism operates through information degradation rather than direct deterrence. Agents self-censor, degraded messages fail to communicate shared discontent, and common knowledge never forms even though the scalar second-order elicitation barely moves. This is consistent with Kuran's (1991) preference-falsification logic while remaining agnostic about whether the full latent belief state is literally unchanged.

Related literature. The paper connects three strands. The global game literature—from the generic uniqueness result of Carlsson and van Damme (1993) through the regime change application of Morris and Shin (2003) and the dynamic extensions of Angeletos et al. (2007) and Chassang (2010)—provides the theoretical framework. The key insight of Morris and Shin (2002), that public signals are overweighted because they predict others' actions, underlies the information design results. Laboratory experiments by Heinemann et al. (2004) confirmed threshold behavior; Szkup and Trevino (2020) found level-k thinking rather than BNE, motivating the cognitive-hierarchy comparison.[1]

The Bayesian persuasion framework (Kamenica and Gentzkow, 2011), applied to regime change by Goldstein and Huang (2016) and to optimal censorship by Kolotilin et al. (2022), provides the information design theory. The political economy of authoritarian control—Edmond (2013) on propaganda, Kuran (1991) on preference falsification, Shadmehr and Bernhardt (2011) on collective action—motivates the surveillance experiments. Empirical evidence that censorship targets coordination (King et al., 2013), surveillance suppresses expression (Penney, 2016; Stoycheff, 2016), and social media enables protest (Enikolopov et al., 2020) provides the real-world stakes.

On the methodological side, Horton (2023) proposed LLMs as "homo silicus." Subsequent work found LLMs struggle in coordination games (Akata et al., 2025; Sun et al., 2025), with model scale not predicting strategic performance (Petrov et al., 2025). Huang et al. (2024) and Carlini et al. (2025) document that alignment shifts risk preferences, motivating narrative rather than tabular payoffs. Critical reviews by Larooij and Törnberg (2025a,b) identify validation challenges that I address through cross-generator robustness (Appendix F.9), scramble/flip falsification, and out-of-sample replication.

The paper proceeds as follows. Section 2 presents the theoretical framework. Section 3 describes the experimental design. Part I (Sections 4–5) establishes that LLM agents implement equilibrium-consistent threshold policies and that pre-play communication creates an exploitable information channel. Part II (Section 6) shows how an information designer can reshape coordination outcomes through Bayesian persuasion, with channel decomposition in Appendix E. Part III (Section 7) tests whether an authoritarian regime can exploit the communication channel through surveillance and propaganda.

2 The Global Game of Regime Change

A continuum of citizens indexed by $i \in [0, 1]$ simultaneously choose whether to join an uprising ($a_i = 1$) or stay home ($a_i = 0$). The regime has strength $\theta \in \mathbb{R}$, drawn from a diffuse (improper uniform) prior. States $\theta \leq 0$ represent regimes so weak they fall without opposition; states $\theta \geq 1$ represent regimes that survive even unanimous attack. The regime falls if the mass of citizens who join exceeds θ (Eq. 1):

$$\text{Regime falls} \iff A \equiv \int_0^1 a_i \, di > \theta. \tag{1}$$

Payoffs depend on the citizen's action and the outcome

[1] Concurrent work by Ruano and Rajan (2026) places LLM depositors in a bank-run coordination game with network contagion, finding sharp phase transitions consistent with Diamond–Dybvig.

(Eq. 2):

$$u_i(a_i, A, \theta) = \begin{cases} B & \text{if } a_i = 1 \text{ and } A > \theta \\ -C & \text{if } a_i = 1 \text{ and } A \leq \theta \\ 0 & \text{if } a_i = 0 \end{cases} \tag{2}$$

where $B > 0$ is the payoff to joining a successful uprising and $C > 0$ is the cost of joining a failed attempt. Non-participants receive zero regardless of the outcome.

Each citizen observes a private signal $x_i = \theta + \varepsilon_i$, where $\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$ independently across citizens.

Proposition 1 (Morris and Shin, 2003)**.** *In the limit of diffuse priors, there exists a unique Bayesian Nash equilibrium in threshold strategies. An agent joins if and only if $x_i < x^*$ (Eq. 3), where*

$$x^* = \theta^* + \sigma\Phi^{-1}(\theta^*) \tag{3}$$

and $\theta^* = B/(B + C)$.

The *attack mass* $A(\theta)$—the theoretical fraction of the population that joins at regime strength θ—is given by:

$$A(\theta) = \Phi\left(\frac{x^* - \theta}{\sigma}\right). \tag{4}$$

This is a decreasing function of θ: weaker regimes face larger uprisings.

I define $J(\theta)$ as the *empirical join fraction*—the finite-sample, observed counterpart to the theoretical attack mass $A(\theta)$. Proposition 1 applies to a continuum of players with diffuse priors. My experiment uses $n = 25$ agents with a normal prior $\theta \sim \mathcal{N}(\bar{z}, 1)$, and agents do not observe (B, C, σ) directly. For finite n the equilibrium cutoff differs from the continuum limit $\theta^* = B/(B + C)$, but the discrepancy shrinks rapidly in n and σ (Morris and Shin, 2003); at $n = 25$ the finite-n correction is small relative to the estimation uncertainty in empirical cutoffs. I therefore compare $J(\theta)$ against the continuum $A(\theta)$ throughout, and validate this choice empirically in Appendix F.6 (Table A15).

An information designer controls the mapping $\pi : \Theta \to \Delta(\mathcal{S})$ from states to signal distributions, but cannot control agents' actions. In my implementation, π is a deterministic mapping from regime strength θ to the parameters of the briefing generator—a system that produces a natural-language intelligence briefing from a z-score derived from the agent's private signal. All stochasticity enters downstream through the LLM's decoding.

The briefing generator has three control parameters: clarity (the width of the Gaussian kernel mapping z-scores to text, where wider kernels produce more ambiguous briefings), directional precision (the slope of the mapping from z-score to briefing sentiment, where steeper slopes produce more accurate signal reflection), and dissent framing (the floor on the probability that the briefing includes language about public discontent).

The designer concentrates manipulation near θ^* using a Gaussian proximity weight (Eq. 5):

$$w(\theta) = \exp\left(-\left(\frac{\theta - \theta^*}{h}\right)^2\right) \tag{5}$$

where $h = 0.15$ is the baseline bandwidth parameter. Each design specifies multiplicative factors for clarity width and directional slope (effective param = base$\times[1+w(\theta)(f-1)]$), and an absolute blend target for the dissent floor (effective = base $\cdot\,(1 - w)$ + target $\cdot\, w$). At $\theta = \theta^*$ the full design is applied ($w = 1$); away from θ^* the parameters revert smoothly to baseline.

The framework generates testable predictions for both the baseline game and information design.

Proposition 2 (Stability design suppresses coordination)**.** *Let the designer widen the clarity kernel by $\delta > 0$ near θ^*, so that briefings near the equilibrium cutoff become less informative. If the empirical join curve is left-shifted ($\hat{\theta}^* < \theta^*$), pooling near θ^* compresses more of the high-joining region than the low-joining region, and the average join fraction falls: $\mathbb{E}[J_{stability}(\theta)] < \mathbb{E}[J_{baseline}(\theta)]$.*

Intuition. Ambiguity near θ^* pools weak- and strong-regime signals. Because the baseline sigmoid is left-shifted ($\hat{\theta}^* < \theta^*$), pooling pulls down join rates in the already-high-joining region ($\theta < \theta^*$) more than it boosts them in the low-joining region, yielding a net decrease.

Proposition 3 (Upper censorship suppresses joining in weak states)**.** *"Upper censorship" refers to concealing unfavorable (weak-regime) states, i.e., the lower tail of θ; the label follows the convention that the upper bound of the censored region is θ^*. If the designer pools all $\theta \leq \theta^*$ to a neutral signal (replacing each agent's private z-score with $z = 0$), the join rate in the pooled region converges to a common plateau J_0—the rate induced by the neutral briefing—which lies below the uncensored join rate for low θ, since $J_0 < J(\theta)$ whenever the uncensored signal would strongly indicate regime weakness. For $\theta > \theta^*$, signals are unchanged. Hence (i) the action–$A(\theta)$ correlation is attenuated, and (ii) the average join fraction falls.*

Intuition. An agent who would join with probability 0.90 at $\theta = 0.20$ under the baseline now receives a pooled signal indistinguishable from $\theta = 0.45$ and joins with probability ≈ 0.50. The "plateau" in Figure 9 is this pooling at work: all agents in the censored region see the same neutral briefing regardless of the true state, collapsing the within-region variation.

Proposition 4 (Surveillance induces a belief–action wedge)**.** *Suppose agents self-censor their communications when monitored. If messages carry information about private signals ($R^2_{msg} > 0$ under baseline), then surveillance degrades message quality ($R^2_{msg} \to 0$) and downstream join rates fall by $\Delta_{surv} < 0$, even though the decision prompt contains no reference to surveillance.*

Intuition. Self-censored messages remove the informational content that communication added. Other agents update on uninformative messages as if they were uninformative, producing the same directional shift as an ambiguity increase—but through the endogenous channel of strategic communication rather than exogenous signal manipulation.

Table A1 in Appendix A reports the eight pre-specified hypotheses and their test results.

3 Experimental Design

The experiment has three parts. Part I tests whether LLM agents implement threshold-like policies in the global game when private signals are conveyed in natural language: a pure treatment (private signals only), a communication treatment (pre-play messaging), and falsification tests. Part II takes the behavioral foundation as given and studies information design: stability/instability designs, censorship, public signal injection, and single-channel decomposition. Part III tests whether an authoritarian regime can exploit the communication channel through surveillance, propaganda, and their interaction. All LLM interactions use the same prompt structure across models.

A note on the state variable. In the theory (Section 2), θ is an unbounded fundamental with special roles for $\theta \leq 0$ (regime falls without opposition) and $\theta \geq 1$ (regime survives even unanimous attack). In Part I experiments, θ is drawn from a normal distribution and agents are not shown payoff parameters (B, C). To evaluate monotonicity on a common scale, I compute the benchmark $A(\theta)$ under the canonical normalization $B = C = 1$ (so $\theta^* = 0.50$) and $\sigma = 0.3$. In Part II, I keep $B = C = 1$ and restrict attention to a fixed θ-grid in $[0.20, 0.80]$ for comparability with the canonical $[0, 1]$ formulation. Table 1 summarizes the three regimes.

Table 1: Experiment regimes across paper sections.

Part	θ distrib.	Baseline	Estimand
I (Align.)	$\mathcal{N}(\bar{z}, 1)$	Pure game	Mean over θ draws
II (Info.)	Grid $[0.2, 0.8]$	Calibrated	Mean over grid
III (Regime)	Same as I	Comm. game	Mean over θ draws

Throughout, $\theta^* = B/(B + C)$ is the theoretical cutoff, $x^* = \theta^* + \sigma\Phi^{-1}(\theta^*)$ the signal cutoff, and $\hat{\theta}^*$ the estimated cutoff from the fitted logistic $P(\text{join} \mid \theta) = 1/(1 + e^{b_0 + b_1\theta})$, where $b_1 > 0$ corresponds to a join curve that is *decreasing* in θ (the expected direction). The cutoff is $\hat{\theta}^* = -b_0/b_1$. I write $\beta \equiv b_1$ for the logistic steepness parameter in tables; larger β means a sharper threshold.

For each country–period, nature draws $\theta \sim \mathcal{N}(\bar{z}, 1)$, where $\bar{z}$ is a public prior mean drawn randomly for each country. Each agent i receives a private signal $x_i = \theta + \varepsilon_i$ and computes a z-score $z_i = (x_i - \bar{z})/\sigma$. The z-score is then translated into a multi-paragraph intelligence briefing by a deterministic generator that maps signal strength to narrative content about regime stability, economic conditions, public sentiment, and coordination prospects. Figure 1 summarizes the signal-to-text-to-decision pipeline.

Agents observe only their private briefing and never the payoff parameters (B, C), the noise level σ, or the prior distribution. The diffuse-prior equilibrium (Proposition 1) serves as the theoretical benchmark, which requires justification. Three considerations support this choice. First, the diffuse-prior BNE is the limit as prior precision vanishes relative to private-signal precision, so it is the standard "uninformative prior" benchmark in the global-games literature. Second, while agents nominally face a hierarchical normal prior ($\theta \mid \bar{z} \sim \mathcal{N}(\bar{z}, 1)$), the briefing generator absorbs $\bar{z}$ through the z-score normalization: agents see only the centered object $z_i = (x_i - \bar{z})/\sigma$, mapped to text and shifted by `cutoff_center`, not θ and $\bar{z}$ separately. The diffuse-prior approximation therefore applies to the *centered private-signal object* governing the briefing, not to a literal structural posterior over the level θ. Third, the paper's claim is behavioral—that the *pattern* of LLM responses matches the BNE prediction—rather than structural: I do not assume agents derive the equilibrium from payoff introspection; the benchmark is the functional form against which behavior is evaluated.

Because the briefing generator maps z-scores monotonically to narrative, any sentiment-extracting model will produce a positive correlation with $A(\theta)$. The empirical contribution lies in the *quantitative structure*: sigmoid shape, payoff-narrative sensitivity, and robustness across seven models. Within-briefing falsification tests (Appendix F.6) confirm that the signal is distributed across all eight evidence domains.

Calibration adjusts a single parameter—the cutoff center—via a damped iterative procedure. The sigmoid shape is emergent and never optimized. Holdout validation confirms no overfitting; 6 uncalibrated models produce $|r| > 0.71$ (Appendix Table A7). Stakes are embedded in the narrative rather than an explicit payoff table, since numeric payoffs caused models to ignore briefing content.

Part I has four core treatments: a *pure global game* (independent decisions), *communication* (pre-play messaging on a Watts–Strogatz network, $k = 4$, $p = 0.3$), *scramble* (cross-period briefing permutation breaking the θ-to-signal link), and *flip* (negated z-scores). Additional identification treatments include an eight-condition *narrative cost ladder*, three alternative *direction transforms* (tanh, linear, step), a *domain ablation* removing coordination- or state-language domains, and a *degraded-messages* control replacing LLM messages with generic text. Part II implements information designs: *stability* (increased ambiguity near θ^*), *instability* (sharpened signals), *public signal injection*, *upper/lower censorship* (Kolotilin et al., 2022), and a *publicness* $\times$ *authority* factorial. Part III tests authoritarian instruments: *surveillance* (monitoring warning in communication prompt) and *propaganda* ($m = 2, 5, 10$ regime plants).

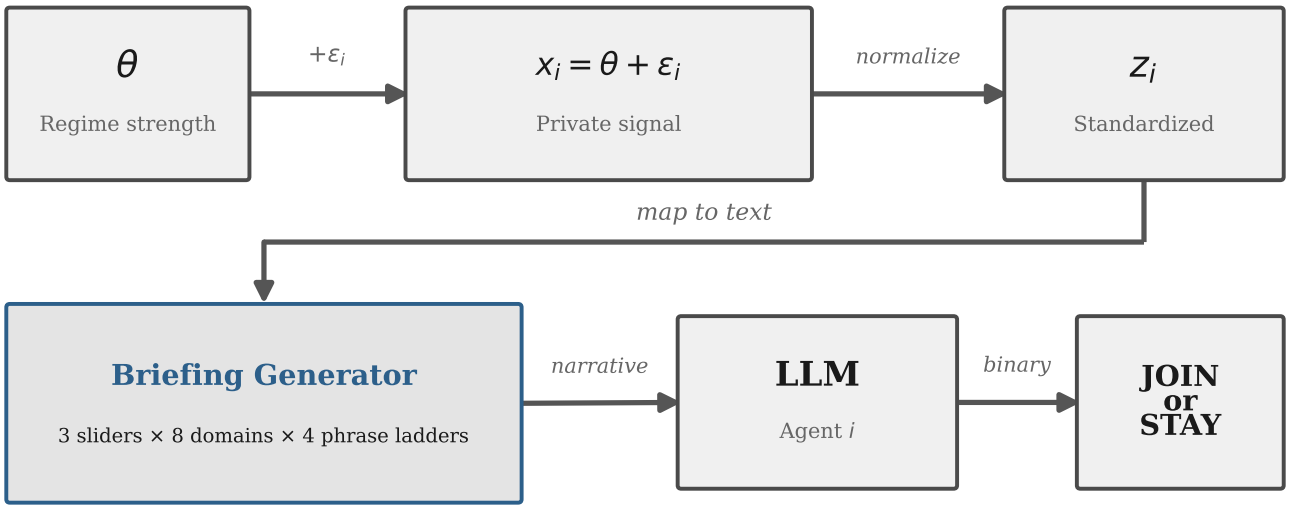


Figure 1: Signal-to-text-to-decision pipeline. Regime strength θ generates private signals x_i, converted to z-scores and rendered into natural-language briefings via 8 evidence domains and 3 latent sliders. The briefing layer is deterministic; all stochasticity enters through LLM decoding.

Table 2: Model summary. Columns report country-period counts in the pure, communication, and falsification (scramble+flip) suites. All runs use $N = 25$ agents per period and $\sigma = 0.3$.

Model	Arch.	Pure	Comm	Falsif.
Mistral Small Creative	Mistral	1000	1000	300
Llama 3.3 70B	Llama	100	100	200
Qwen3 30B	Qwen (MoE)	100	100	200
GPT-OSS 120B	GPT	200	200	1000
Qwen3 235B	Qwen (MoE)	200	200	200
Trinity Large	Arcee	100	100	200
MiniMax M2-Her	MiniMax	100	100	200
Total		**1800**	**1800**	**2300**

Notes: 7 models spanning 6 architecture families. N = country-periods with 25 agents each. All runs use $\sigma = 0.3$ and temperature = 0.7 unless otherwise stated.

I test seven models spanning six families (Table 2). All experiments use $n = 25$ agents per country–period and $\sigma = 0.3$. Part I benchmarks use $B = C = 1$ ($\theta^* = 0.50$); payoff comparative statics vary narrative stakes. All LLM calls use temperature = 0.7 with a single sample per decision, routed through OpenRouter and cached by request hash (Appendix G). Throughout, n denotes agents per country–period and N denotes the number of country–periods.

The unit of randomization is the country–period (θ draw plus agent-level decoding stochasticity). For agent-level regressions, standard errors are clustered at the country–period level to account for within-period correlation among agents sharing the same θ. For period-level correlations, I report Fisher-z confidence intervals. The 25 agents within a period share the same θ and calibration; their decisions are conditionally independent given their private signals.[2]

For the information design experiments, I fix $B = C = 1$ (so $\theta^* = 0.50$) and a 9-point θ-grid in $[0.20, 0.80]$, with 30–60 repetitions per (design, θ) cell and 25 agents each. The primary model is Mistral Small Creative; cross-model replication uses four additional models.

Table 3 maps each treatment to the theoretical channel it tests, the directional prediction, and the observed result.

Of the eight pre-specified hypotheses, five survive Bonferroni correction at $\alpha = 0.05/8$: alignment (H1), flip (H3), communication (H4, paired $t = 6.383$), censorship (H6), and surveillance (H7). H2 (scramble) confirms a null prediction by design. H5 (stability) moves in the predicted direction but is statistically ambiguous ($p = 0.408$). H8 (propaganda) is directionally correct but non-significant ($d = -0.049$; post-hoc power = 0.145, MDE at 80% power = 6.3 pp). Exploratory analyses are reported with uncorrected p-values.

4 Results

Result 1 (Threshold-Policy Alignment)**.** *Across seven models and 1,800 country–periods in the pure global game treatment, the Pearson correlation between the empirical join fraction and the theoretical attack mass $A(\theta)$ (Eq. 4)*

[2] I do not claim literal conditional independence—shared prompt structure and model weights introduce common factors. The clustering accounts for within-period correlation.

Table 3: Treatment map. Each row describes a treatment, the channel it tests, the unit of observation, the theoretical prediction, and the observed result. Part I tests whether LLM agents play the global game; Part II studies information design; Part III tests authoritarian exploitation of the communication channel. r denotes Pearson correlation; Δ is the change in mean join rate relative to the relevant baseline (percentage points).

Treatment	Part	Channel tested	Unit	Prediction	Observed
Core treatments					
Pure global game	I	Baseline monotonicity	Period	$r(J, A(\theta)) > 0$	$r = +0.80$
Communication	I	Pre-play messaging	Period	Small within-cell effect	+2.10 pp (paired)
Scramble (cross-period)	I	Content vs. format	Period	$r \to 0$	$r = +0.03$
Flip	I	Signal direction	Period	Sign reversal	$r = -0.80$
B/C narratives	I	Payoff comparative statics	Period	Centering + payoff sensitivity	Grid recenters; header swing 57.6 pp
Information design (Part II, primary model)					
Stability	II	Ambiguity near θ^*	Period	Small $\downarrow$ join	-1.4 pp (n.s.)
Instability	II	Sharpened signals	Period	Sharper sorting; mean sign ambiguous	+4.8 pp
Public signal	II	Common knowledge channel	Period	$\downarrow$ join (dominates private)	-39.2 pp
Censor (upper)	II	Pooling weak states	Period	Plateau below θ^*	-15.4 pp
Censor (lower)	II	Pooling strong states	Period	Plateau above θ^*	+3.7 pp
Authoritarian instruments (Part III)					
Surveillance	III	Preference falsification	Period	$\downarrow$ join	-13.5 pp
Propaganda (m=10)	III	Information contamination	Period	$\downarrow$ join, saturating	-2.3 pp (behavioral)
Within-briefing falsification					
Observation shuffle	I	Bullet ordering vs. content	Period	r unchanged	$r = +0.912$
Domain scramble (coord.)	I	Coordination-relevant domains	Period	$\lvert r\rvert$ falls if domains drive signal	$r = +0.921$
Domain scramble (state)	I	State-capacity domains	Period	$\lvert r\rvert$ falls if domains drive signal	$r = +0.928$

Notes: $r = r(J, A(\theta))$: Pearson correlation between empirical join fraction and theoretical attack mass. Δ: change vs. relevant baseline (pp). Part I results are pooled across 7 models; Parts II–III use the primary model (Mistral Small Creative).

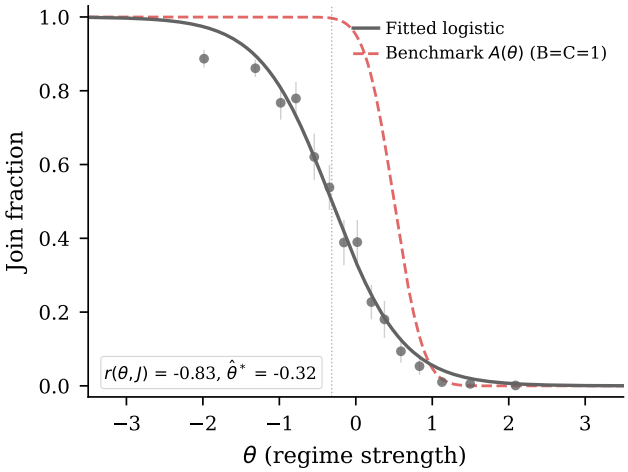


Figure 2: Empirical join fraction vs. θ (Mistral, 1,000 periods). The empirical sigmoid is shifted leftward ($\hat{\theta}^* = -0.32$) relative to theory ($\theta^* = 0.50$). Cross-model results in Table 4 (mean $r = +0.80$, all $p < 0.001$).

averages $r = +0.80$ *(*$p < 0.001$ *for every model).*[3]

Figure 2 shows the empirical sigmoid for the primary model. Correlations range from $r = +0.75$ (Mistral) to $r = +0.84$ (Trinity), with the pooled OLS: $J = 0.13 + 0.54\,A(\theta)$ ($R^2 = 0.48$). The slope below unity is expected: narrative mediation compresses the effective signal range relative to the theoretical attack mass. All seven per-model correlations remain significant at $p < 0.001$ under country-clustered inference.

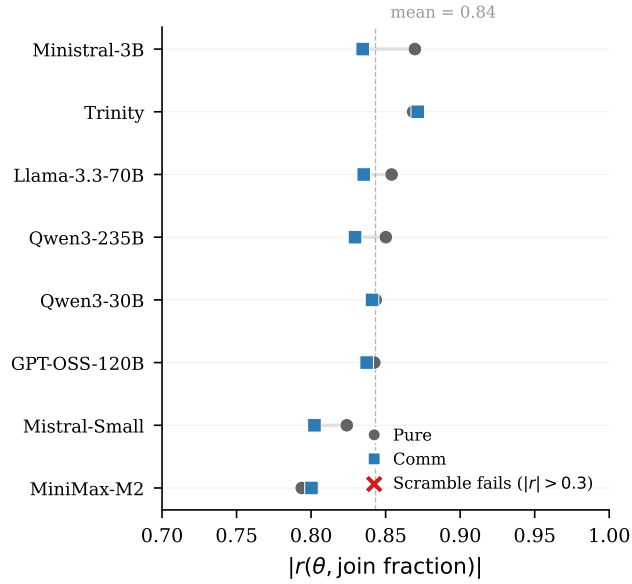


Figure 3: Cross-model signal monotonicity. Points report $r(J, A(\theta))$ under pure and communication.

Alignment is stable across architectures (Figure 3): correlations span $r \in [+0.75, +0.84]$ despite parameter counts spanning an order of magnitude. Mean join rates vary, reflecting model-specific action biases that shift the intercept but not the correlation. Estimated cutoffs vary ($\hat{\theta}^* \in [-0.32, +0.10]$); the disciplined evidence on cutoff

[3]The primary model (Mistral) was run in multiple batches with `--append`, producing 1,000 rows from 680 unique (country, period, θ) configurations. Averaging within each configuration (deduplication) changes r from 0.753 to 0.739; all qualitative results are unaffected.

Table 4: Threshold-policy alignment by model and treatment. Cells report Pearson r between the empirical join fraction and the benchmark attack mass $A(\theta)$ under $B = C = 1$ (so $\theta^* = 0.50$); 95% Fisher-z confidence intervals in brackets for main treatments.

	Main treatments		Falsification			
Model	Pure	Comm	Scramble	Flip	n_{pure}	Mean join
Mistral Small Creative	+0.75 [0.72, 0.78]	+0.74 [0.71, 0.76]	−0.03	−0.81	1000	0.387
Llama 3.3 70B	+0.82 [0.74, 0.87]	+0.78 [0.69, 0.85]	+0.02	−0.84	100	0.439
Qwen3 30B	+0.76 [0.66, 0.83]	+0.77 [0.68, 0.84]	+0.13	−0.87	100	0.499
GPT-OSS 120B	+0.79 [0.74, 0.84]	+0.78 [0.72, 0.83]	−0.01	−0.80	200	0.407
Qwen3 235B	+0.80 [0.74, 0.84]	+0.76 [0.70, 0.81]	+0.06	−0.85	200	0.416
Trinity Large	+0.84 [0.77, 0.89]	+0.83 [0.75, 0.88]	+0.05	−0.80	100	0.464
MiniMax M2-Her	+0.82 [0.74, 0.87]	+0.82 [0.74, 0.87]	+0.01	−0.62	100	0.436
Pooled	+0.76 [0.74, 0.78]	+0.75 [0.73, 0.77]	+0.01	−0.79	1800	0.409
Mean across models	+0.80	+0.78	+0.03	−0.80	—	—

Notes: $r = r(J, A(\theta))$: Pearson correlation between empirical join fraction and theoretical attack mass under $B = C = 1$. 95% Fisher-z confidence intervals in brackets. Join fraction uses valid decisions only (excluding parse errors). Pooled: all country-periods concatenated; Mean: equal-weighted average across models.

location comes from comparative statics, where changing narrative stakes shifts $\hat{\theta}^*$ in lockstep with payoff theory (Table 6, Figure 4). Taken together with the fixed-briefing narrative ladder, these comparative statics show that the behavior is more sophisticated than the level-k patterns typically observed in human laboratory subjects, without resting the claim on the shifted-grid sweep alone.

Human subjects in coordination game experiments consistently exhibit level-k cognitive hierarchy behavior rather than full equilibrium reasoning (Szkup and Trevino, 2020). To test whether LLM agents play the BNE or a level-k heuristic, I evaluate theoretical attack masses under BNE, L_1 (agents best-respond believing others randomize uniformly, i.e. L_0 randomizes with probability $\frac{1}{2}$ at each signal), and L_2 (agents best-respond believing others play L_1). The three benchmarks predict different threshold cutoffs θ^*_{BNE}, $\theta^*_{L_1}$, $\theta^*_{L_2}$ for each (B, C) configuration; while these thresholds coincide when $B = C$ (yielding $\theta^* = 0.50$ for all three), they diverge under asymmetric payoffs. Because agents never observe (B, C) directly, the identification operates on the *text-signal* channel: varying the payoff ratio shifts the θ-grid and thus the briefings agents receive, and I compare the observed cutoff shifts against the three theoretical predictions (Table 5). Across the seven payoff conditions, the fitted cutoff tracks the BNE benchmark more closely than L_1 or L_2 (RMSE = 0.199, $r = +0.74$ vs. 0.307/+0.45 and 0.348/+0.35), but the shifted-grid design means this evidence is best read as a centering diagnostic rather than a standalone rejection of level-k behavior. The cleaner evidence against a pure level-k reading comes from the fixed-briefing narrative ladder and the broader robustness checks above.

The claim is behavioral: agents do not observe (B, C), σ, or n, yet the *pattern* of responses—monotone sigmoid, narrative-sensitive cutoff, and beliefs that strongly predict decisions—is robust across architectures and falsifiable. Three identification strategies establish that the regularity reflects more than text classification.

Stakes identification. Two complementary tests address how agents respond to strategic stakes, but only one changes text while holding the briefing body fixed.

First, a *payoff sweep* varies the game-theoretic (B, C) across seven configurations (Table 5). In each condition, agents receive standard briefings with no explicit cost/benefit language—changing (B, C) shifts θ^* and the θ-grid, which in turn shifts the z-score centering point $\bar{z} = \theta^*$. Agents never observe (B, C) directly. The seven fitted cutoffs $\hat{\theta}^*$ track $\theta^* = B/(B + C)$ with $r = 0.997$ ($p < 0.001$; Figure 4). An important caveat: because both the θ-grid and $\bar{z}$ shift with θ^*, the distribution of z-scores—and hence the distribution of briefing texts—is held approximately constant across conditions. In z-units, empirical join rates are nearly identical across all seven conditions (within 1–2 pp at each relative grid position). The payoff sweep therefore tests the end-to-end pipeline's θ^*-centering property—confirming that the calibrated signal-to-text-to-decision chain correctly translates theoretical cutoff shifts into behavioral cutoff shifts—rather than demonstrating that agents infer payoff magnitudes from unchanged text. There is no same-briefing/different-payoff manipulation in this design, so the sweep should be read as a centering diagnostic, not as standalone evidence of direct payoff reasoning. The sharper test of strategic-stakes framing comes from the explicit narrative conditions below, where the context text *itself* changes.

Second, an *eight-condition narrative cost ladder* tests whether qualitative strategic-stakes framing shifts behavior. A context header—varying from "existential consequences" through "ambiguous stakes" to "zero personal cost"—is prepended to otherwise identical briefings (Table 5, bottom panel). Agents never observe payoff parameters; the briefing body is unchanged across conditions. The ladder produces a clean monotonic response (Figure 4b): mean joining rises from 21.3% (very high cost) through

Table 5: Cost/benefit conditions. *Top:* payoff sweep—only (B, C) vary. *Bottom:* narrative cost ladder—only a prepended header varies.

Condition	B	C	Narrative framing
Payoff sweep (standard briefings, no header):			
$\theta^* = 0.25$	0.33	1	—
$\theta^* = 0.50$	1.00	1	—
$\theta^* = 0.75$	3.00	1	—
(+ four intermediate values; see text)			
Narrative cost ladder (B=C=1; header prepended):			
Very high cost	1	1	Disappearances, families targeted
Moderate high	1	1	Prison sentences, job loss
Neutral	1	1	Consequences uncertain
Moderate low	1	1	Brief questioning, released
Very low cost	1	1	Guaranteed asylum, zero cost
Controls:			
Placebo	1	1	Weather information (irrelevant)
High cost v2	1	1	Paraphrase of very high cost
Low cost v2	1	1	Paraphrase of very low cost

Notes: Payoff sweep: $B = \theta^*/(1-\theta^*)$, C=1. Z-score center and θ-grid shift with θ^*; briefing text distribution is approximately constant. Narrative ladder: B=C=1; only the header varies. Join rates: very high cost 21.3%, moderate high 46.6%, neutral 46.7%, moderate low 68.4%, very low 78.9%. Placebo 32.7%, v2 variants 27.4%/63.8%. Full headers in Appendix A.

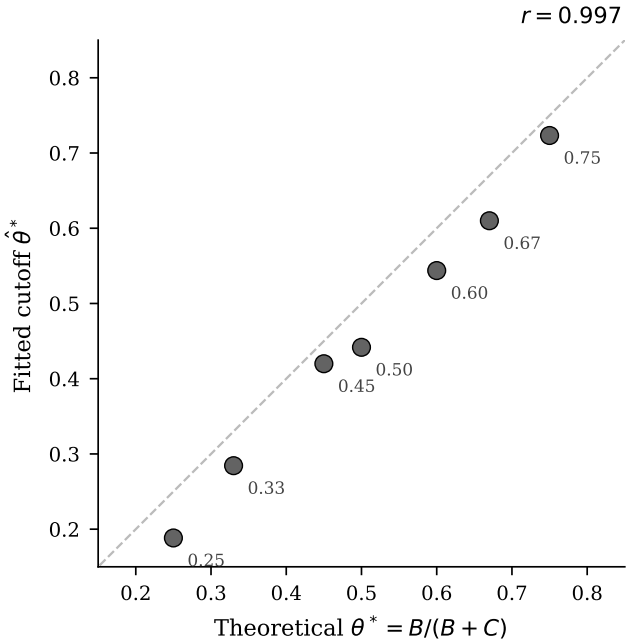


Figure 4: Fitted cutoff $\hat{\theta}^*$ vs. theoretical θ^* across seven B/C ratios ($r = 0.997$). All conditions use standard briefings; the z-score grid shifts with θ^*, so this tests pipeline centering rather than direct payoff sensitivity (see text).

32.7% (placebo) and 46.7% (neutral) to 78.9% (very low cost)—a 57.6 pp range. Paraphrase variants of the extreme conditions track their originals (27.4% vs. 21.3% for high-cost; 63.8% vs. 78.9% for low-cost), confirming robustness to surface wording. A placebo header containing only weather information produces 32.7% joining—below the 40.9% baseline, consistent with irrelevant context diluting the briefing signal. The briefing-text classifier (Appendix C) conditions only on the briefing body and cannot see the narrative header, confirming that the briefing text alone is insufficient to reproduce the shift. I therefore interpret the ladder as evidence that agents respond strongly to strategic-stakes framing embedded in context, not as a clean decomposition between literal payoff reasoning and every other high-level framing cue. An intensity-matched nonstrategic placebo would be needed for that narrower claim. The narrative cost ladder nevertheless provides the paper's sharpest fixed-briefing identification: eight conditions, monotonic dose-response, paraphrase robustness, and a placebo control, all with identical briefing content.

Belief reports. Stated probability judgments are strongly associated with action (Pseudo $R^2 = 0.975$; Table A16). Beliefs track the theoretical success probability ($r = +0.78$) and predict actions beyond what signals alone explain (partial $r = +0.64$), but the near-perfect pseudo-R^2 should be read as predictive association rather than proof of mediation because these beliefs are elicited post-decision.

Classifier baselines. Text features alone cannot reproduce LLM behavior. A TF-IDF logistic classifier achieves 88.4% accuracy in-distribution—top features are regime-stability language ("trajectory," "deterioration" predicting JOIN; "stability," "absence" predicting STAY)—but fails on the surveillance chilling effect (+13.0 pp gap; Table A3 in Appendix C). Additional identification tests are reported in Appendix C.

The correlation is invariant to decoding temperature (for the primary model, $r \in [+0.76, +0.79]$ across $T \in \{0.3, 0.7, 1.0\}$; full three-model sweep in Appendix F.5).

Generator robustness. Two tests address the concern that the briefing generator's architecture mechanically produces the sigmoid. First, replacing the logistic function mapping z-scores to directional sentiment with three alternatives—tanh, piecewise linear, and a hard step function at $z = 0$—yields nearly identical correlations: $r = 0.83$ (tanh), $r = 0.83$ (linear), $r = 0.84$ (step), compared with $r = +0.75$ (logistic baseline). The sigmoid emerges from the LLM's processing of natural language, not from the particular functional form of the signal-to-text mapping. Second, a domain ablation test removes subsets of the briefing generator's eight evidence domains. Removing the two coordination-language domains (public coordination climate and dissent signals) raises mean joining by 2.8 pp while preserving the threshold relationship almost exactly ($r(J, A(\theta)) = 0.90$). Removing the four regime-state domains (elite cohesion, security posture, economic fundamentals, institutional resilience) lowers mean joining by 3.4 pp and attenuates the threshold relationship

Table 6: Cost/benefit narrative comparative statics. High cost: narrative emphasizes severe reprisals for failed action. Low cost: narrative emphasizes minimal consequences. Theory predicts higher perceived cost lowers the cutoff (less joining).

Design	N	Mean join	$r(J, A(\theta))$	Cutoff $\hat{\theta}^*$ (SE)	Δ (pp)
Baseline	270	0.409	+0.88	0.39 (0.007)	—
High cost	270	0.190	+0.87	0.13 (0.010)	-21.9
Low cost	270	0.693	+0.88	0.72 (0.007)	+28.4

Notes: Primary model, information design θ-grid. Identical briefings across conditions; only the benefit/cost narrative framing varies. Δ: change vs. baseline mean join (pp).

($r(J, A(\theta)) = 0.86$). The mechanism claim therefore rests on the preserved-versus-attenuated sigmoid diagnostics, not on the small difference between the two mean shifts alone.

What matters for Part II is that the behavioral regularity is robust enough to serve as a platform for studying how information structures shift coordination outcomes.

Result 2 (Signal Dependence)**.** *Cross-period scrambling of briefings reduces the mean within-country correlation from $r = +0.80$ to $r = +0.03$ across seven models. The pooled correlation drops from $r = +0.76$ to $r = +0.01$ (Fisher $z = 26.17$, $p < 0.001$).*

The scramble preserves the marginal distribution of briefing content but breaks the mapping from each period's θ to the signals agents receive.[4]

Result 3 (Signal Direction)**.** *Inverting the signal direction flips the mean correlation from $r = +0.80$ to $r = -0.80$ across seven models. The pooled correlation moves from $r = +0.76$ to $r = -0.79$ (Fisher $z = 53.85$, $p < 0.001$).*

The pure → scramble → flip pattern replicates across all seven models (Figure 5; Table 4).

A text-only baseline ($\hat{p}_{\text{join}} = 1 - d$, where d is the briefing's internal direction score) correlates with LLM decisions at $r = 0.79$, but the LLM's join curve is substantially steeper (Figure 6), producing threshold-like behavior rather than linearly tracking tone.

5 Communication

Result 4 (Communication opens an information channel with a small positive effect)**.** *Pre-play communication changes the mean join rate by -0.4 pp when averaging per-model means (equal-weighted across the seven models). Three estimators quantify the effect at different levels of aggregation:*

1. *Model-average: compute each model's mean join under pure and communication, then average the seven per-model deltas. This yields -0.4 pp—near zero because some models shift positively and others negatively.*
2. *Matched pairs: merge pure and communication periods on the full task key (model, country, period, θ, and the associated signal cell), yielding $n = 1479$ matched cells. The within-cell difference is $\Delta = +2.10$ pp ($t = 6.383$, $p < 0.001$), revealing a small but reliably positive effect once compositional differences are absorbed.*
3. *Pooled unpaired: pool all pure and communication periods (observation-weighted, so models with more periods receive more weight). The difference is +1.23 pp ($p = 0.327$), not significant, because cross-θ variation swamps the small treatment effect.*

The point-estimate discrepancy is primarily a weighting issue, not just a precision issue. The matched estimator equal-weights the 1479 model–country–period–θ task cells, whereas the pooled unpaired estimator weights every period by raw observation count and therefore lets high-volume models dominate the average. The corrected pooled match drops only a negligible number of cells because of missing communication outcomes, so the difference between +2.10 pp and +1.23 pp is overwhelmingly reweighting rather than support loss.

Communication preserves the signal structure ($r = +0.78$ vs. $+0.80$ under pure; Figure 7). The effect on join rates is heterogeneous across models and near zero on average, consistent with passive Bayesian updating concentrated in weak-regime environments.

How informative are the messages? Regressing θ on message features yields $R^2 = 0.12$—messages carry real information, noisily. This informativeness is what makes the channel exploitable: surveillance degrades R^2 to 0.02 (Section 7), an 81% reduction. Communication creates the information topology that authoritarian instruments exploit. With group-size knowledge, the communication premium reverses (-3.4 pp; Appendix F).

A *degraded-messages* control confirms that the communication effect operates through message content, not the mere structure of a communication round. I replace all LLM-generated messages with generic uninformative text ("I'm still assessing the situation," "It's hard to say what will happen") while preserving the two-stage protocol and omitting any surveillance framing. Degraded messages drop the mean join rate from 41.1% to 20.2%—well below even the pure game (38.7%), indicating that uninformative messages actively dampen coordination by flooding the channel with noise. This result has two implications: first, the communication benefit in the main treatment is driven by *informative* content, not by the opportunity to communicate per se; second, any mechanism that degrades message quality—including self-censorship under surveillance—should produce a comparable suppression.

[4] Because the cross-period permutation operates within countries, all scramble correlations use within-country (country-demeaned) Pearson r.

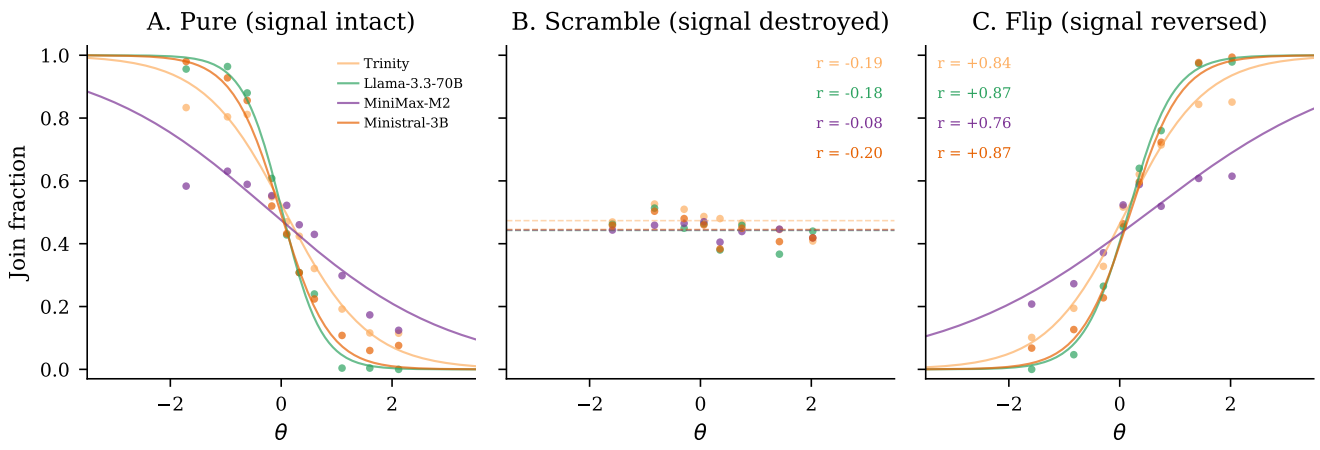


Figure 5: Falsification triptych. *Left:* Pure (mean $r = +0.80$). *Center:* Scramble ($r = +0.03$). *Right:* Flip ($r = -0.80$).

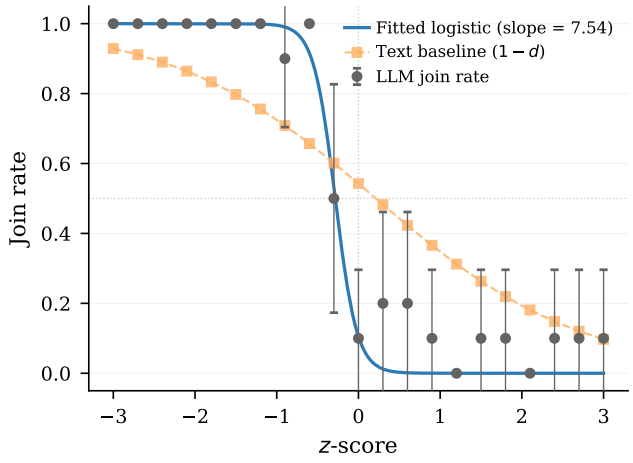


Figure 6: Text baseline test. The LLM join curve is steeper than the naïve text predictor $(1 - d)$, indicating processing beyond surface sentiment.

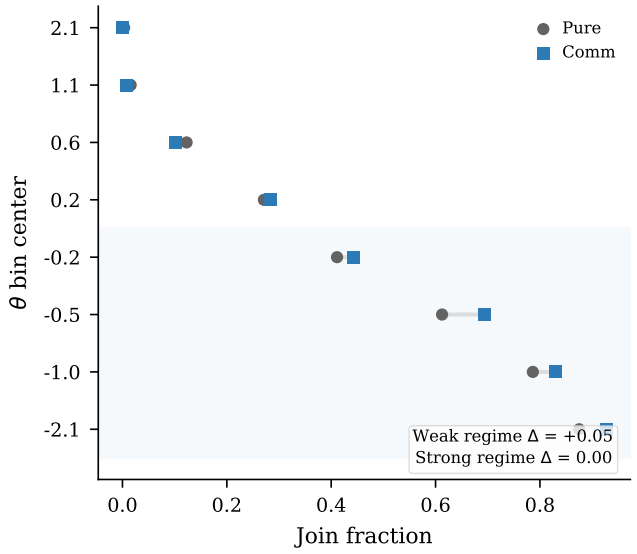


Figure 7: Communication effect by regime strength, pooled across seven models.

6 Information Design

Each information design modifies the briefing generator's slider functions near θ^* while agents' decision process remains unchanged, testing whether theoretical predictions (Goldstein and Huang, 2016; Kolotilin et al., 2022) produce the predicted shifts when agents are LLMs.

A conceptual clarification: these interventions are *implemented* as modifications to the briefing generator's parameters (clarity width, directional slope, dissent floor), which alter the statistical relationship between θ and the text signal. They are not information-structure transformations in the strict Blackwell sense—one cannot compute a likelihood ratio between treatment and baseline signal distributions when the signal space is natural language. Rather, they are the closest feasible analog: holding the decision environment fixed and varying the *mapping from states to signals* through the generator's parameters. The theoretical predictions (Propositions 2–3) are stated in terms of these generator-level manipulations, not abstract signal structures. The empirical question is whether the directional predictions survive the text-to-decision pipeline.

Figure 8 displays the design effects; Table 7 summarizes the main results. The baseline condition produces a mean join rate of 40.9%($r = +0.885$, $p < 0.001$).

Result 5 (Information Design Can Shift Coordination)**.** *The largest information-design treatments produce sizable shifts in coordination relative to baseline; the bundled stability design moves in the predicted direction but is modest.*

Consistent with Proposition 2, the stability design suppresses coordination: mean join falls from 40.9%to 39.5% (-1.4 pp), and the regime-fall rate drops from 43.3% to

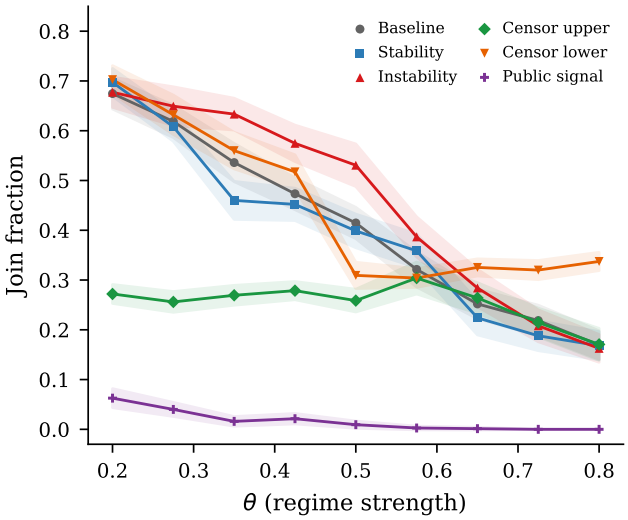


Figure 8: Join fraction vs. θ under six information designs (Mistral Small Creative).

41.9%. But this bundled effect is small: the pre-specified H5 test is statistically ambiguous ($p = 0.408$), so I treat stability as directional evidence rather than a clean rejection. The instability design, by contrast, *raises* mean joining to 45.6% (+4.8 pp). The average sign is positive because sharpening the briefing increases joining at eight of the nine θ grid points, especially in the intermediate weak-state region, and only slightly lowers joining at the strongest state. With the estimated behavioral cutoff left of θ^*, sharper signals push many mid-range states further into the JOIN region rather than toward STAY. The instability design therefore sharpens sorting, but in this calibrated finite-grid environment the net effect is to increase average joining.

The public signal injection is the strongest common-knowledge treatment in the paper: besides the header, each agent receives an appended public bulletin generated from four shared observations of the true state. This produces the largest reduction (1.7%, -39.2 pp). The theoretical interpretation (Morris and Shin, 2002) is that agents overweight public signals because such signals predict others' beliefs, crowding out private information. The framing-only appendix checks are milder. In the publicness × authority factorial and Table A2, the standard private briefing body is unchanged and only the source/publicness header is varied. Those designs therefore probe framing around a common signal, not the full public-bulletin treatment, which is why their join rates remain in the 35–44% range rather than collapsing to 1.7%. I therefore treat the main public-bulletin treatment as a bundled public-information intervention rather than as a design that cleanly orthogonalizes common knowledge from authority or provenance cues. Within the milder header-only family, an alternative account—that the header acts as an authoritative instruction cue rather than a common-knowledge

Table 7: Information design treatment summary (primary model: Mistral Small Creative). r is the Pearson correlation $r(J, A(\theta))$ between join fraction and theoretical attack mass.

Design	Mean	r	Δ (pp)	N
Baseline	0.409	+0.88	—	270
Stability	0.395	+0.87	-1.4	270
Instability	0.456	+0.87	+4.8	270
Public signal	0.017	+0.53	-39.2	540
Scramble	0.432	+0.11	+2.3	270
Flip	0.663	−0.82	+25.5	270

Notes: Data from the primary model (pure treatment; $\theta \in [0.20, 0.80]$ on a 9-point grid; N=25 agents per period). Mean join uses valid decisions; r is Pearson $r(J, A(\theta))$ across rep-level periods.

statistic—is partially testable. A publicness × authority factorial crosses two dimensions: whether the briefing is framed as distributed to all citizens (public) or provided individually (private), and whether it is attributed to the Ministry of Internal Affairs (authority) or an anonymous source. The results separate the two channels within that header-only design: authority framing increases joining under public framing (40.8% vs. 35.4%, +5.4 pp) but produces no significant effect under private framing (38.1% vs. 39.8%, −1.7 pp). This interaction—authority matters only when others also observe the signal—is consistent with strategic complementarity: authority adds credibility that agents use to predict others' responses, not just to update their own beliefs. The "authority alone" account predicts a main effect regardless of publicness, which is not observed.

Treatment effects by θ are shown in Figure A4 (Appendix D).

Kolotilin et al. (2022) proved that the optimal information structure is upper censorship—one-sided pooling that conceals unfavorable states. Following that convention, "upper" refers to censoring all weak-regime states up to the upper bound θ^* of the pooled region. Upper censorship therefore pools weak-regime states ($\theta \leq \theta^*$) into a neutral signal; lower censorship pools strong-regime states.

Result 6 (Upper Censorship Suppresses Joining in Weak States)**.** *As predicted by Proposition 3, upper censorship lowers the mean join rate to 25.4%(-15.4 pp vs. baseline) and attenuates the action–attack-mass correlation (r = +0.271). The effect is concentrated in the censored region ($\theta \leq \theta^*$), where weak-regime states are pooled to a neutral briefing and join rates flatten.*

Pooling generates a flat join-rate "plateau" in the censored region (Figure 9): when agents cannot distinguish $\theta = 0.20$ from $\theta = 0.50$, they behave as if the regime is borderline rather than clearly weak.

A common-knowledge test makes the censorship rule transparent. When agents are told "regime censors are suppressing unfavorable intelligence above a certain severity

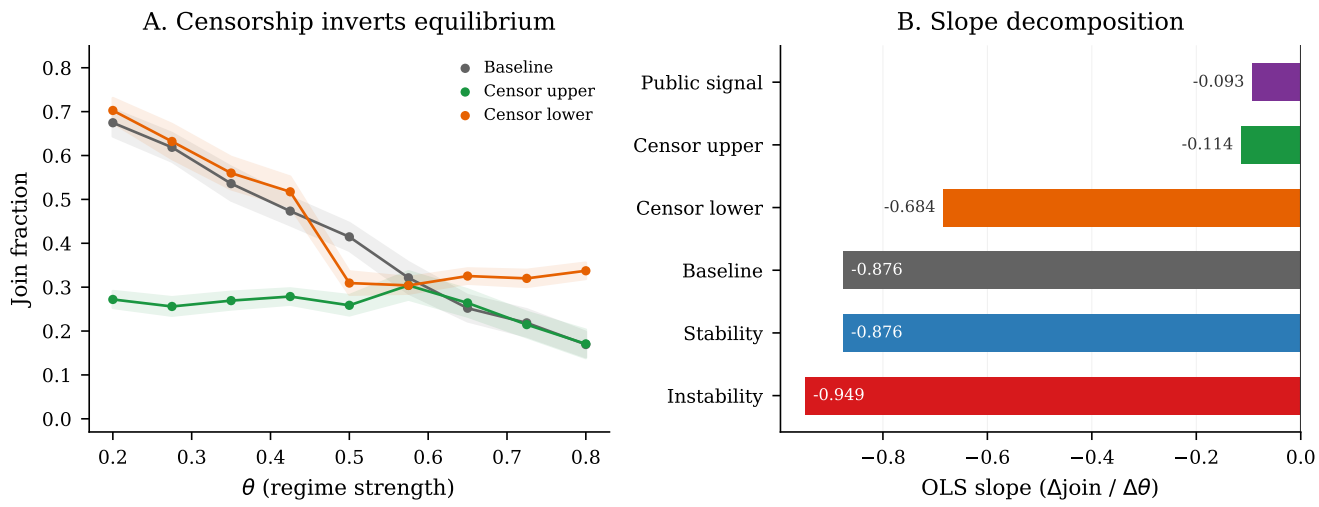


Figure 9: Censorship effects. *Left:* Upper censorship creates a flat plateau; lower censorship attenuates the slope and raises mean joining. *Right:* OLS slope decomposition.

threshold," the mean join rate returns close to the uncensored baseline (43.2% vs. 40.9% baseline; Table A12)—common knowledge of the censorship rule largely neutralizes the pooling distortion, consistent with theory assuming informed receivers (Kolotilin et al., 2022). The naïve censorship result extends the theory to receivers who fail to account for the information structure—a realistic scenario where the censorship apparatus is opaque.

Result 7 (Lower Censorship Attenuates Comparative Statics)**.** *Lower censorship produces a mean join rate of 44.5%(+3.7 pp vs. baseline) and attenuates the action–attack-mass correlation ($r = +0.798$ vs. baseline $r = +0.885$). Below θ^*, signals are unchanged and agents see their true (weak-regime) briefings. Above θ^*, pooling strong-regime states to a neutral briefing ($z = 0$) raises join rates—agents who would otherwise see discouraging intelligence now receive uninformative briefings and default toward joining. The resulting step-up at θ^* compresses the join-rate range across the grid: the censored region (high θ) rises to moderate joining while the uncensored region (low θ) retains its baseline rates, flattening the overall slope and reducing $|r|$.*

The attenuation is consistent with pure Bayesian pooling: censoring one tail compresses the signal range, reducing $|r|$ without inverting the sign. The mechanism operates through the replacement of each censored agent's z-score with $z = 0$, which on the calibrated generator (cutoff center ≈ -0.32) produces a briefing that reads as mildly optimistic about regime stability. For the censored grid points ($\theta \geq \theta^*$), this uniform "mildly stable" briefing generates join rates substantially higher than the near-zero baseline rates at those θ values, while the uncensored grid points ($\theta < \theta^*$) retain their normal—and highly variable—join rates. The net effect is a step-up in joining at θ^* that compresses the slope without inverting it.

Falsification within information design confirms signal dependence: scrambling collapses the correlation to $r = +0.107$ ($p = 0.08$); flipping inverts it to $r = -0.820$ ($p < 0.001$). A harder scramble generating all briefings from $\theta = 0.5$ produces $r \approx 0$ for both Mistral and Llama (Figure A5 and bandwidth sensitivity in Appendix D).

7 Surveillance

Kuran (1991) labeled the mechanism underlying authoritarian stability "preference falsification"; I adopt the more operationally precise term *belief–action wedge* because my experimental design measures the gap between stated beliefs and expressed actions rather than hidden preferences *per se*.

In the surveillance treatment, the communication prompt is augmented with a warning that communications are being monitored by regime security services. The manipulation affects only the communication phase; the decision prompt is unchanged. The isolation is architectural: each LLM call is stateless—the decision-stage API call receives only the briefing and the (now self-censored) messages, with no memory of the communication-stage surveillance warning. Any difference in join rates must therefore operate through the *content* of the messages that the decision stage receives.

This stateless architecture is essential for causal interpretation. Because the decision prompt contains no surveillance reference and the LLM has no persistent memory across calls, the behavioral wedge cannot arise from agents "remembering" they were surveilled. Instead, the chain is: (1) surveillance warning → (2) self-censored messages (less informative, less action-oriented) → (3) decision stage receives degraded messages → (4) agents who would have been persuaded by authentic messages now

lack the coordination signal and default toward STAY.

Two placebo variants—monitored for research (no consequences) and aggregated anonymously—produce no significant deviation from the communication baseline (+2.6 pp and +4.0 pp, both $p > 0.2$; Appendix F.6), confirming that the chilling effect is specific to the surveillance warning's strategic content. The degraded-messages control (Section 5) provides independent confirmation of the indirect channel: exogenously replacing messages with generic text—without any surveillance framing—drops joining to 20.2%, a larger suppression than surveillance itself (27.6%). This is the expected pattern if surveillance operates *primarily* by degrading message informativeness: the exogenous control removes all information, while self-censorship under surveillance only partially degrades it.

Result 8 (Surveillance Produces a Large Chilling Effect)**.** *Consistent with Proposition 4, surveillance reduces mean join rates in the primary model (Mistral Small Creative) from 41.1% to 27.6%, a shift of* -13.5 *pp* ($p < 0.001$)*. The action–attack-mass correlation is preserved* ($r = +0.68$ *vs.* $r = +0.74$ *under the communication baseline), indicating that surveillance operates as a level shift rather than disrupting signal processing.*

Surveillance contaminates the information environment with self-censored messages, pushing join rates well below the communication baseline (Figure 10). The effect replicates across five models: Mistral (−13.5 pp), GPT-OSS (−11.9 pp), Qwen3 30B (−10.9 pp), Llama (−8.9 pp), and Qwen3 235B (−3.3 pp), averaging −9.7 pp.

Indirect channel: self-censored messages degrade information. Surveillance shifts stated beliefs by only −0.7 pp ($p = 0.18$, not significant) while shifting join rates by −12.9 pp ($p < 0.001$). A decomposition formalizes this (Eq. 6):

$$\underbrace{\Delta\bar{J}}_{\text{total}} = \underbrace{\frac{\partial \bar{J}}{\partial b}\,\Delta\bar{b}}_{\text{belief updating}} + \underbrace{\Delta\bar{J}\big|_{\bar{b}\text{ fixed}}}_{\text{belief–action wedge}} \tag{6}$$

At the level of the scalar elicitation, virtually the entire effect passes through the belief–action wedge—agents change what they *do* without changing what they *state they believe*. The "belief" elicited is the agent's assessment of uprising success probability, which tracks θ through the briefing. Because the briefing is unchanged across conditions and beliefs are elicited from the same private signal, stated first-order beliefs remain stable. Second-order beliefs—the elicited estimate of what fraction of others will join—also fail to shift (31.0% → 30.9%, $p = 0.5866$). I therefore treat these scalar belief measures as coarse descriptive readouts, not as a separately validated mediator for communication-quality shocks.

How can agents exhibit a large action shift with no shift in either first- or second-order stated beliefs? Three (non-exclusive) explanations apply. First, point-estimate elicitation may miss changes in *precision*: agents could hold the same mean belief about others' behavior but with greater uncertainty under surveillance, and the decision heuristic may be risk-averse (defaulting toward STAY when uncertain). Second, the decision and belief elicitations are separate stateless LLM calls; the decision call processes the actual (degraded) messages in its context window, while the belief call may not fully reflect the strategic update that shaped the choice. Third, the mechanism may operate through higher-order reasoning—"others are also self-censoring, but I cannot verify this"—that a single-number elicitation does not capture. Elicited punishment risk is ≈8.0/10 across all conditions with negligible differences between JOIN and STAY agents (Table A20), supporting the contaminated-information interpretation over a direct deterrence channel.

The R^2 of message content on true θ drops from 0.12 to 0.02—an 81% reduction in informational content.

Direct channel: decision-stage suppression. A separate hold-fixed test isolates the direct decision-stage channel by replaying baseline messages verbatim (holding received messages fixed) while adding a surveillance warning to the *decision* prompt ("your actions may be monitored"). Join rates shift by −20.9 pp ($p < 0.001$), confirming that surveillance can operate through a direct decision-stage suppression channel when the warning is present at the action stage. Crucially, this channel is absent in the main surveillance design, where the decision prompt contains no surveillance reference—the main treatment isolates the *indirect* (communication-poisoning) channel. The hold-fixed test thus serves as a boundary condition: it shows what surveillance *could* achieve if the regime also warned at the action stage, but it does not contaminate the main estimate, which captures only the communication degradation mechanism. I do not estimate the stacked condition in which agents both receive self-censored messages and simultaneously see a surveillance warning at decision time; that combined treatment would answer a different question about additive channels rather than the estimand of interest here.

Message content confirms self-censorship: "act" drops from 15.8% to 4.1% of messages under surveillance; "collapse" drops from 9.6% to 0.6%. Among agents who privately decide to JOIN, 61.0% send action-signaling messages under surveillance vs. 80.0% under regular communication.

Instrument interactions. When propaganda ($m = 5$) and surveillance are combined, the mean join rate among real citizens falls to 24.3%, a shift of −16.8 pp from the communication baseline (41.1%). The sum of individual effects is −15.5 pp (surveillance −13.5 pp + propaganda −2.0 pp), so the combined effect (−16.8 pp) is approximately additive: once surveillance has suppressed expressed dissent, propaganda adds only modest additional deterrence.

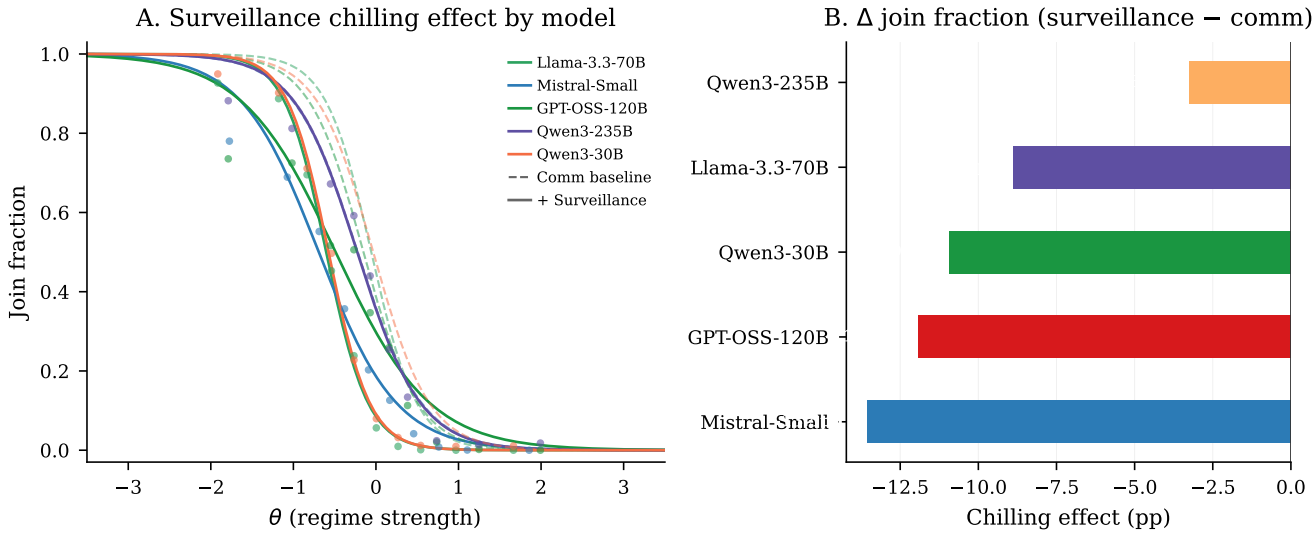


Figure 10: Surveillance reduces join rates across five models, averaging −9.7 pp.

Surveillance and censorship, by contrast, show a complementary interaction (Table 8): the marginal effect of surveillance in absolute terms is larger under censorship (−11 to −14 pp) than at baseline (−2 pp). Much of this difference is mechanical: with a 3% baseline, the floor compresses the absolute effect. On the log-odds scale, the surveillance effect is roughly 25–30% larger under censorship (Δ log-odds ≈ -1.5 vs. -1.2), indicating a modest complementarity that persists after accounting for the floor. Table 8 uses the *communication information-design grid*—a fixed θ-grid $[0.20, 0.80]$ with communication and calibrated briefings, the same regime used for Part II. The low baseline join rate (3.0%) in this regime, compared with the 40.9% pure-game baseline on the same θ-grid (Table 7), reflects the communication channel: message exchange amplifies the majority tendency at each θ-cell. On this grid the calibrated sigmoid yields below-50% joining at most grid points, so the exchanged messages predominantly signal caution, creating a herding equilibrium at STAY that suppresses joining far below the pure-game rate. The key estimand in Table 8 is the relative surveillance *increment* Δ across censorship conditions, not the absolute level. Censorship degrades private signal quality, making the communication channel relatively more important, and surveillance poisons that channel. With both instruments active, agents have neither private signals to trust nor authentic messages to learn from (Figure 11). The censorship effect under surveillance varies across architectures (Table 9): the marginal impact of censorship given surveillance depends on model-specific resolution of pooled private signals and self-censored messages.

Table 8: Surveillance × censorship interaction in the communication game (primary model: Mistral Small Creative).

Design	No Surv.	Surv.	Δ (pp)
Baseline	0.030	0.009	-2.1
Upper cens.	0.151	0.037	-11.4
Lower cens.	0.177	0.042	-13.5

Notes: Both columns use the communication information-design grid ($\theta \in [0.20, 0.80]$, 9 points, calibrated briefings, communication game). The low baseline level (3.0%) reflects this grid regime, not the Part I normal-draw regime (≈40%); see text for discussion. "Surv." adds surveillance during messaging. All entries are means of `join_fraction_valid`. Δ is the surveillance increment (Surv. − No Surv.) in percentage points.

Table 9: Cross-model surveillance × censorship interaction. Δ: change vs. surveilled baseline (pp).

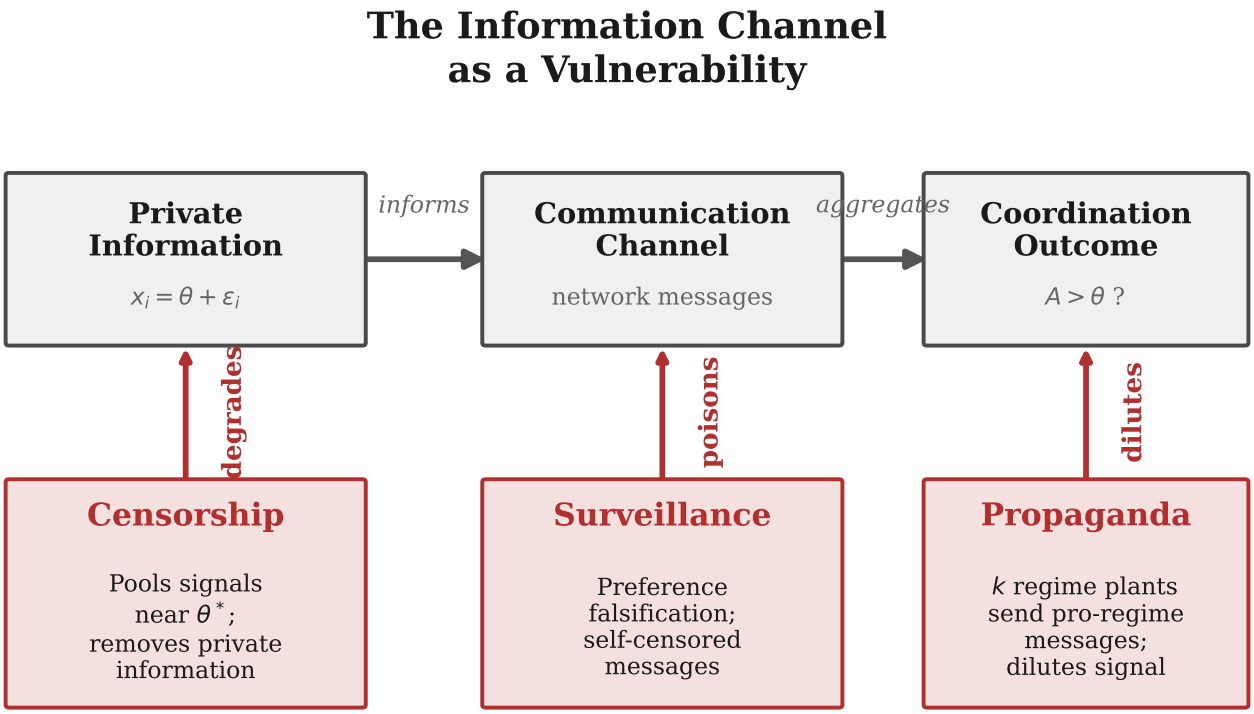
	Mean join (surv.)			Δ (pp)	
Model	Base	Upper	Lower	Δ_U	Δ_L
Mistral	0.009	0.037	0.042	+2.8	+3.3
Llama 70B	0.114	0.039	0.115	-7.5	+0.1
GPT-OSS	0.316	0.177	0.312	-13.9	-0.4
Qwen 235B	0.336	0.321	0.468	-1.5	+13.1

Notes: Communication treatment, information design θ-grid, surveillance active during messaging. Δ: change vs. surveilled baseline (pp). $N = 25$ agents per period.

7.1 Propaganda

Edmond (2013) modeled propaganda as the regime shifting citizens' signal distributions. I introduce m propaganda agents (regime plants, always STAY) into the 25-citizen communication network. The *overall* join rate includes plants; the *real citizen* rate isolates the behavioral effect (Figure 12).

Result 9 (Propaganda Suppresses Coordination Primarily Through Mechanical Dilution)**.** *Mean join fraction (including plants) falls from 41.1% ($m = 0$) to 37.5%*



Instrument Interactions

Surv. + Propaganda:

Approximately additive

Surv. + Censorship:

Super-additive: -11.4 pp under censorship vs -2.1 pp at baseline

The regime does not need to change what citizens believe; it needs only to make them uncertain about each other.

Figure 11: How authoritarian instruments attack the coordination chain. Surveillance and censorship are complements; propaganda saturates quickly.

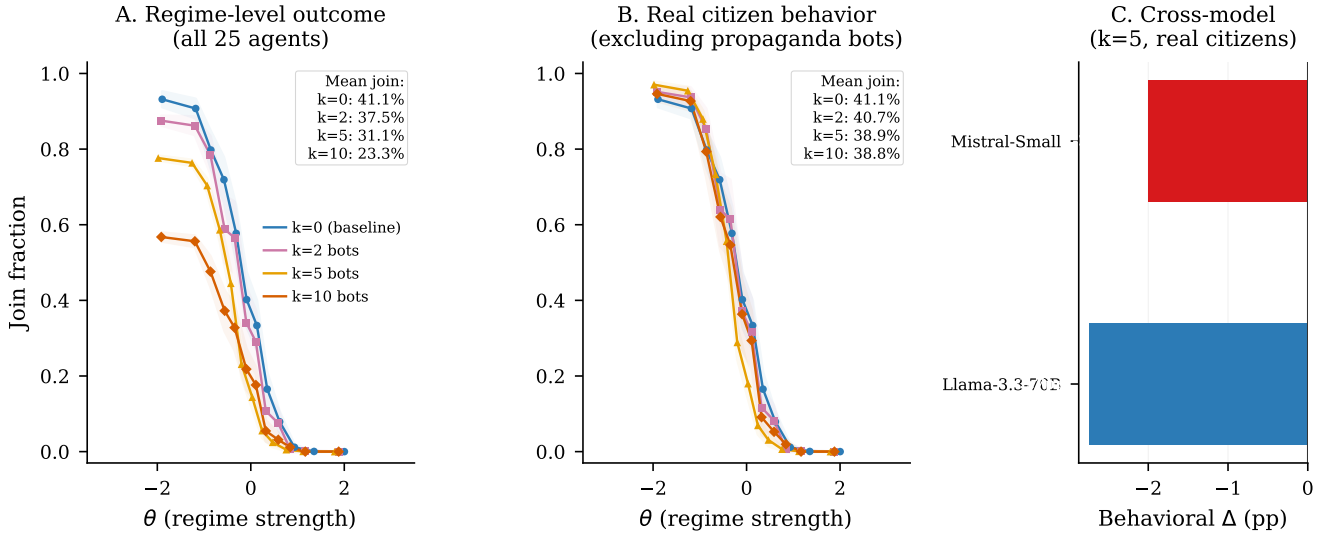


Figure 12: Propaganda dose-response: Mistral (primary) and Llama (replication).

($m = 2$), 31.3% ($m = 5$), and 23.3% ($m = 10$). However, the behavioral effect on real citizens is much smaller and saturates: 41.1% ($m = 0$), 40.7% ($m = 2$, -0.4 pp), 39.1% ($m = 5$, -2.0 pp), 38.8% ($m = 10$, -2.3 pp).

The mechanical channel (plants always STAY) is approximately linear in m; the behavioral channel saturates quickly—doubling plants from 5 to 10 produces no additional effect (-0.3 pp, $p = 0.6676$). The pattern replicates with Llama (-2.7 pp at $m = 5$).

Message content reveals the mechanism (Table 10). Propaganda injects regime-loyal vocabulary ("loyal" rises from 1.5% to 11.4% at $m = 10$) while coordination language declines ("ready" drops from 27.1% to 18.5%). Among real agents who STAY, caution-coded messages rise from 51.7% to 54.7%—agents echo propaganda. Among those who JOIN, action signaling remains stable at $\approx$85.1%.

Table 10: Propaganda and surveillance effects (primary model: Mistral Small Creative). "All" includes propaganda agents; "Real" excludes them (computed from logs). Δ is the change in real-agent mean join vs. baseline communication.

	Mean join			
Treatment	All	Real	r	Δ (pp)
Comm (baseline)	0.411	0.411	+0.74	—
Prop $m = 2$	0.375	0.407	+0.72	-0.4
Prop $m = 5$	0.313	0.391	+0.72	-2.0
Prop $m = 10$	0.233	0.388	+0.72	-2.3
Surveillance	0.276	0.276	+0.68	-13.5
Prop+Surv	0.194	—	+0.61	—

Notes: Primary model (Mistral Small Creative), communication treatment. "All" includes propaganda plant agents; "Real" excludes them. Δ: change in real-agent mean join vs. baseline communication (pp).

8 Conclusion

LLM agents implement threshold policies closely consistent with the unique Bayesian Nash equilibrium, and this behavioral regularity converts the experiment into a testbed for information design. The central result is that the same communication channel enabling coordination also enables its suppression: surveillance induces self-censorship that propagates through the network as degraded, less action-oriented messages. A degraded-messages control confirms this is an information-content mechanism: exogenously replacing messages with uninformative text—without surveillance framing—produces an even larger suppression. The channel that solves the coordination problem also transmits the regime's poison. Censorship and surveillance are complements—degrading private signals makes the communication channel more important, and surveillance poisons precisely that channel.

The identification strategy rests on multiple independent tests. The sigmoid relationship between regime strength and join rates is robust to the functional form of the signal-to-text mapping (logistic, tanh, linear, and step functions all yield $r \approx 0.83$). Domain ablation confirms that state-strength evidence, not coordination-language cues, drives the behavioral pattern. An eight-condition narrative cost ladder produces a 57.6 pp monotonic swing with paraphrase robustness and a placebo control, identifying sensitivity to strategic-stakes framing under fixed briefing content even though it does not isolate every possible framing channel. A publicness $\times$ authority factorial shows common-knowledge logic within the header-only framing checks, even though the main public-bulletin treatment remains bundled with source/provenance cues. These

tests collectively establish that the behavioral regularity reflects genuine information processing, not mechanical text classification.

A rough welfare accounting illustrates the stakes. Under uniform $\theta \in [0, 1]$, the BNE yields per-citizen expected payoff of 0.27 (on a $[-1, 1]$ scale), roughly half the 0.50 achievable if all citizens could coordinate on the equilibrium threshold θ^* (the true first-best is 1.0, since coordinated attack succeeds for all $\theta < 1$; the gap between 0.27 and 0.50 isolates the cost of signal noise). Surveillance reduces citizen welfare by a further 14%—regime survival rises from 50% to 54%—while the stability design produces a comparable 13% welfare loss. These calculations are stylized (they treat the observed level shifts as if they were uniform across θ), but they quantify the core asymmetry: the regime's instruments improve its survival at citizens' expense, and the welfare cost of each instrument exceeds its effect on average join rates because the chilling effect concentrates in states where coordination is most valuable.

Three limitations warrant discussion. First, whether the underlying mechanism is approximate Bayesian reasoning or a learned heuristic that happens to match BNE remains open. The behavioral claim—that the pattern is robust, replicable, and falsifiable—does not require resolving this question, but interpreting the welfare implications requires it. Second, the experiments are static; real coordination problems are dynamic (Angeletos et al., 2007; Chassang, 2010), and the dynamics of belief updating under persistent surveillance may differ from the one-shot effects documented here. Third, external validity remains the fundamental challenge for "homo silicus" (Horton, 2023): what LLM agents do in a coordination game is evidence about the structure of the game, not necessarily about how humans would play it. This paper makes no claim that LLMs predict human coordination behavior. What LLMs offer is a calibrated behavioral regularity—the sigmoid—that responds to information design in theoretically predicted ways, creating a testbed for evaluating the *comparative statics* of information structures. The question "does upper censorship suppress coordination more than lower censorship?" can be answered empirically in this framework without requiring that the absolute level of coordination match any human population. The cross-model replication (seven architectures, five surveillance variants) further suggests that the qualitative patterns are properties of the information structure rather than artifacts of particular model training.

Despite these limitations, the framework offers something neither laboratory experiments nor field data can: a setting where the information environment—signal precision, correlation structure, disclosure rules, communication monitoring—can be manipulated while holding everything else fixed. The experimental template is reusable, and the data and code are publicly available.

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A Experimental Design Details

Table A1 reports the eight pre-specified hypotheses, their estimands, test statistics, and outcomes. H1–H4 are tested on pooled data across all seven models (H4 uses a paired t-test matched on model/country/period/θ task cells); H5–H8 use the primary model (Mistral Small Creative). Bonferroni-corrected threshold: $\alpha = 0.00625$. H8 (propaganda dose-response) is directionally correct but underpowered: Cohen's $d = -0.049$, post-hoc power $= 0.145$, and the minimum detectable effect at 80% power is 6.3 pp—the real-citizen effect (-2.0 pp) falls below this threshold.

B Belief Elicitation

Beliefs are elicited on a 0–100 probability scale immediately after each decision ("How likely is it that the regime falls this period?"). Figure A1 documents calibration quality against the theoretical posterior; Figure A2 shows second-order beliefs (predictions about others' join rates) are unchanged by surveillance; Figure A3 confirms monotone signal–belief mapping nonparametrically. Because these are post-decision scalar elicitation calls, I use them descriptively rather than as a separately validated mediator for message-quality shocks.

Beliefs were also elicited under four information design treatments (baseline, stability, instability, censor upper; $n = 6{,}750$ agents per design). Under baseline, JOIN agents state mean belief 64.2 while STAY agents state 35.8—a 28.4 pp wedge consistent with strong belief–action sorting. Under upper censorship, the JOIN–STAY wedge narrows to 15.3 pp (JOIN: 52.7, STAY: 37.4), consistent with pooling compressing the signal range. Under instability, the JOIN–STAY wedge widens to 41.0 pp (JOIN: 71.1, STAY: 30.1), consistent with sharpened signals producing more polarized beliefs. These patterns show that information design shifts stated beliefs in the predicted direction and that those post-decision belief reports continue to track the behavioral response, but they do not by themselves validate the belief instrument for exogenous communication-informativeness shocks.

C Identification Test Details

Table A2 reports OLS and logit estimates for a separate low-intensity common-knowledge framing experiment. Table A3 compares text-classifier accuracy under baseline and surveillance conditions, establishing that the chilling effect is invisible to classifiers restricted to the intelligence briefing body alone. Table A4 repeats the classifier exercise across the cost/benefit sweep.

Narrative header texts. The eight narrative cost ladder conditions prepend the following headers to each agent's standard intelligence briefing (the briefing body is identical across conditions):

Very high cost: "CONTEXT: The regime maintains an extensive network of informants and has a documented history of disappearances following failed uprisings. Participants' families are routinely targeted. The cost of unsuccessful action is existential."

Moderate high cost: "CONTEXT: Participants in past failed uprisings have faced prison sentences of several years and loss of employment. The personal cost of unsuccessful action is significant but not life-threatening."

Neutral: "CONTEXT: The consequences of participating in a failed uprising are uncertain and vary case by case. Some participants have faced penalties while others have not. The personal stakes are ambiguous."

Moderate low cost: "CONTEXT: Recent legal reforms have limited the regime's ability to punish dissent. Past participants faced only brief questioning and were released without charges. The personal risk of action is modest."

Very low cost: "CONTEXT: The international community has guaranteed safe passage and asylum for all participants regardless of outcome. Even failed uprisings carry essentially zero personal cost."

Placebo: "CONTEXT: The country has experienced significant weather disruptions this year, including severe flooding in the eastern provinces and an extended drought in agricultural regions. Infrastructure damage has been substantial."

Paraphrase variants (high cost v2, low cost v2) use different surface wording to express the same cost level. The seven payoff-sweep conditions (Table 5) use no narrative header; agents receive only the standard briefing generated from their private signal.

D Information Design Details

Table A5 reports the realized slider values at the mean signal for the main Part II parameterization ($\theta^* = 0.50$, $\theta \in [0.20, 0.80]$), confirming that the proximity-weighted parameter modulation produces the intended intermediate signal profiles in the replication-relevant design. Stability collapses clarity near θ^* (making briefings ambiguous); instability sharpens it (making briefings more informative). Censorship produces constant slider profiles across the pooled region, as expected from replacing z-scores with 0.

Figure A4 plots $\Delta(\theta)$ — the treatment-induced shift in join rate as a function of regime strength — for each design. Figure A5 confirms internal validity: the scramble and flip falsifications replicate within the information design

Table A1: Pre-specified hypotheses and test results. H1–H4 use pooled Part I data across all seven models; H5–H8 use the primary model (Mistral Small Creative). Effect size: r for correlations (H1–H3), Cohen's d or d_z for mean comparisons (H4–H8). "Supported" at $\alpha = 0.05$.

H	Hypothesis	Test	Stat	p	Effect	Supported?
H1	Sigmoid Response	Pearson	0.757	<0.001	0.757	Yes
H2	Scramble Falsification	Pearson	0.014	0.638	0.014	Yes
H3	Directional Sensitivity	Pearson (1-sided)	-0.791	<0.001	-0.791	Yes
H4	Communication Channel	Paired t	6.383	<0.001	0.166	Yes
H5	Ambiguity Pooling	t-test	-0.828	0.408	-0.071	No (ambiguous)
H6	Censorship Distortion	t-test	-12.207	<0.001	-1.051	Yes
H7	Surveillance Chilling	t-test	-8.277	<0.001	-0.370	Yes
H8	Propaganda Dose-Response	t-test	-0.892	0.372	-0.049	No (ambiguous)

Notes: H1–H4: pooled across all seven models (H4 uses paired t-test matched on model/country/period/θ task cells). H5–H8: primary model (Mistral Small Creative). Bonferroni survivors at $\alpha = 0.00625$: H1, H3, H4, H6, H7. H8 post-hoc power: 0.14 (Cohen's $d = -0.049$; MDE at 80% power = 6.3 pp).

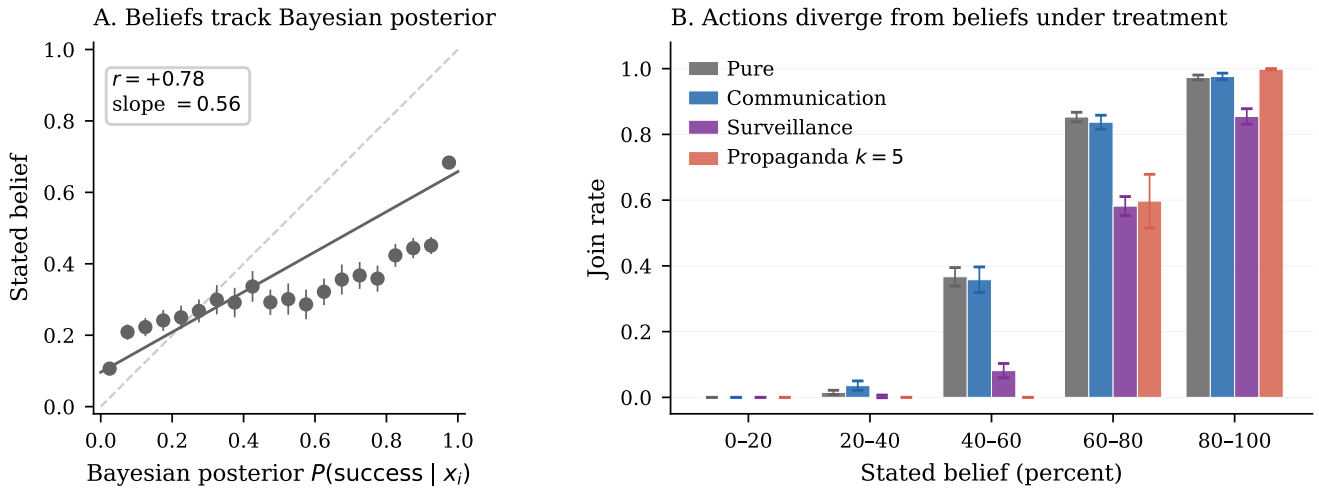


Figure A1: Belief elicitation (Mistral). *Left:* Stated beliefs vs. theoretical $P(\text{success} \mid x_i)$ ($r = +0.78$). *Right:* Join rate by belief bin.

Table A2: Common knowledge × coordination intensity. Each cell reports mean join rate (270 country–periods). The CK main effect is +6.1 pp ($p = 0.0003$); the interaction is -1.2 pp ($p = 0.62$). This is a header-only framing test, not the main public-signal bulletin treatment.

	Low coord	High coord	Δ (coord)
Private framing	37.4%	37.8%	+0.4 pp
CK framing	43.5%	42.7%	-0.8 pp
Δ (CK)	+6.1 pp	+4.9 pp	

Notes: Primary model, information design θ-grid. 270 country-periods per cell. CK framing changes only the publicness/source header ("widely shared"); the private briefing body is otherwise unchanged. High-coord: coordination-cue intensity amplified. Unlike the main public-signal treatment, no separate public bulletin is appended.

Table A3: Classifier baselines. Accuracy and AUC are 5-fold CV on pure-treatment data. "Pred. join (surv.)" is the classifier's predicted join rate when applied to surveillance-treatment briefings; "Actual" is the LLM's observed rate. The gap measures the surveillance wedge invisible to text classifiers.

Classifier	Acc.	AUC	Pred. join (surv.)	Actual (surv.)	Gap
BoW TF-IDF	88.4%	0.947	40.5%	27.6%	13.0 pp
Slider logistic	88.4%	0.941	40.4%	27.6%	12.8 pp
Keyphrase	60.2%	0.594	8.4%	27.6%	-19.2 pp

Notes: Primary model (Mistral Small Creative). Accuracy and AUC from 5-fold cross-validation on pure-treatment data. Gap = classifier-predicted join rate − actual LLM join rate under surveillance (pp).

θ-grid, ruling out confounds from the grid's θ range. The proximity bandwidth h controls how sharply manipulation concentrates around θ^*; the baseline $h = 0.15$ matches the θ-grid spacing and is approximately optimal (Table A10, Appendix F).

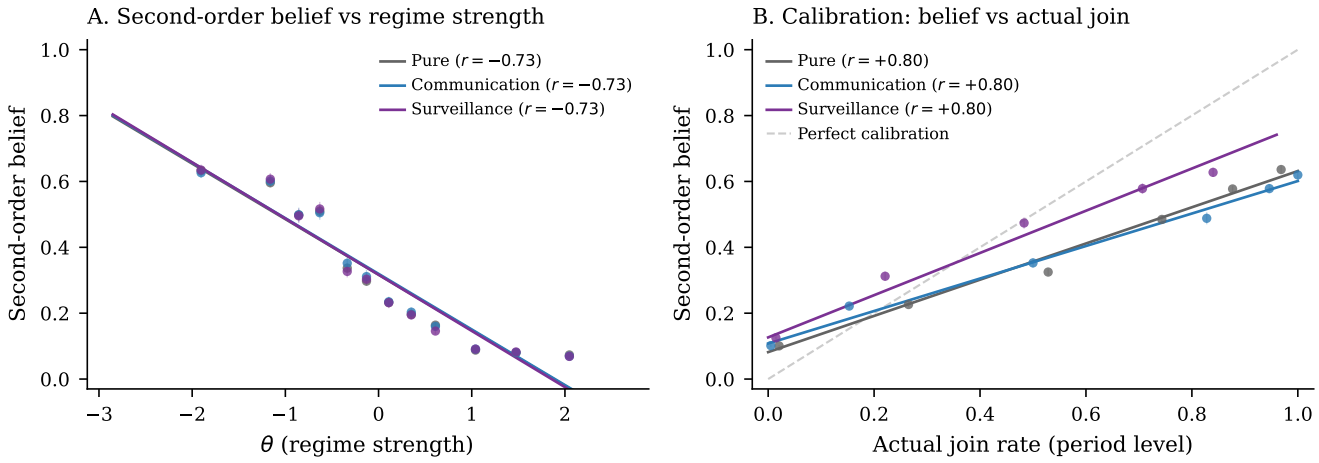


Figure A2: Second-order beliefs. Surveillance overlaps with pure; no significant shift.

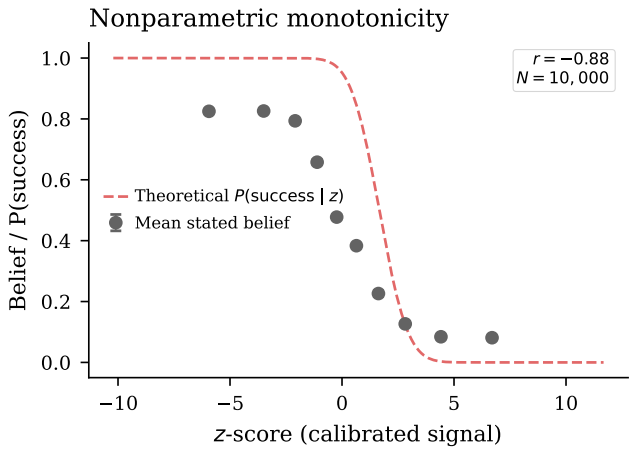


Figure A3: Nonparametric monotonicity: mean belief by z-score bin vs. theoretical benchmark.

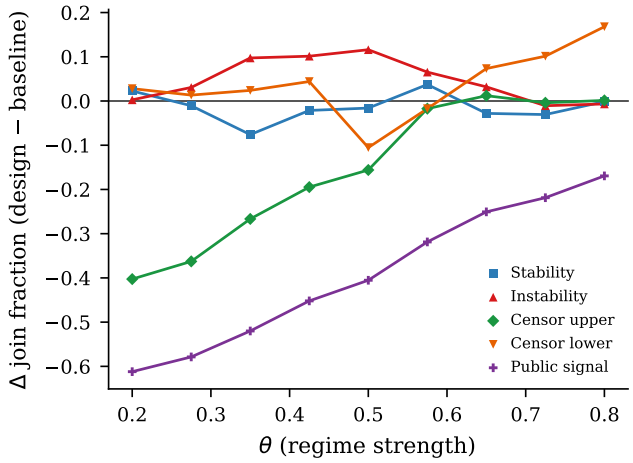


Figure A4: Treatment effect $\Delta(\theta)$ by information design.

Table A4: B/C comparative statics: classifier vs. actual LLM behavior. Actual join rates shift by ≈ 50 pp across payoff conditions; the classifier, trained on text features alone, cannot predict this shift.

Condition	N	Classif. pred.	Actual	Gap (pp)
Baseline ($\theta^* = 0.50$)	270	40.9%	40.9%	+0.0
High cost ($\theta^* = 0.25$)	270	40.9%	19.0%	+21.9
Low cost ($\theta^* = 0.75$)	270	40.9%	69.3%	-28.4

Notes: Primary model (Mistral Small Creative). Logistic classifier trained on baseline slider features. Gap = classifier-predicted join rate − actual LLM join rate (pp).

Table A5: Representative realized slider values by information design and θ in the main Part II parameterization. Direction (D): 0 = regime weak, 1 = strong. Clarity (C): 0 = maximally ambiguous, 1 = clear. Coordination (K): 0 = quiet, 1 = open mobilization. $\theta^* = 0.50$.

	$\theta = 0.20$			$\theta = 0.35$			$\theta = 0.50$			$\theta = 0.65$			$\theta = 0.80$		
Design	D	C	K	D	C	K	D	C	K	D	C	K	D	C	K
Baseline	.30	.64	.66	.38	.51	.59	.47	.43	.53	.56	.43	.46	.64	.53	.39
Stability	.30	.62	.66	.41	.22	.59	.49	.06	.53	.54	.18	.46	.64	.51	.39
Instability	.30	.64	.66	.33	.64	.59	.44	.91	.53	.58	.57	.46	.64	.53	.39
Upper cens.	.46	.04	.53	.46	.04	.53	.46	.04	.53	.56	.43	.46	.64	.53	.39
Lower cens.	.30	.64	.66	.38	.51	.59	.46	.04	.53	.46	.04	.53	.46	.04	.53

Notes: Values are averages of the realized `briefing_mean_*` summaries in the main Part II output files for the primary model. Representative points are taken directly from the replication-relevant grid $\theta \in \{0.20, 0.35, 0.50, 0.65, 0.80\}$. Stability collapses clarity near θ^*; instability sharply increases clarity in the middle of the grid; upper and lower censorship pool the corresponding side of the threshold to a common neutral profile.

D.1 Hard Scramble

A harder scramble test generates *all* briefings from a single fixed state ($\theta = 0.5$), severing any possible correlation between θ and briefing content by construction. Under this hard scramble, $r(J, A(\theta)) = +0.05$ ($p = 0.42$) for Mistral and $r = +0.11$ ($p = 0.08$) for Llama—both indistinguish-

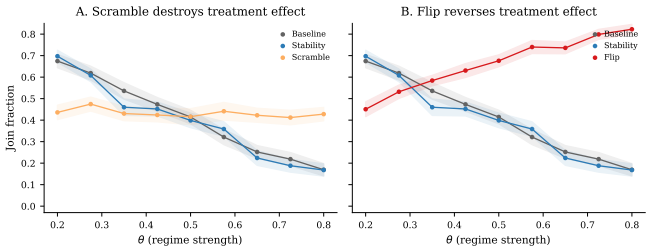


Figure A5: Falsification within information design: scramble ($r = +0.107$), flip ($r = -0.820$).

Table A6: Single-channel decomposition of the stability design (primary model: Mistral Small Creative).

Channel	Mean	r	Δ (pp)
Full stability	0.395	+0.87	-1.4
Clarity only	0.405	+0.89	-0.4
Direction only	0.389	+0.87	-2.0
Dissent only	0.406	+0.88	-0.2
Sum of channels	—	—	-2.6
Full design	—	—	-1.4

Notes: Each row is a separate infodesign run for Mistral Small Creative on the same θ grid as Table 7. Δ reports the mean difference vs. the baseline infodesign mean (Table 7).

able from zero.

E Decomposition: Which Channel Drives the Stability Effect?

The stability design manipulates three channels simultaneously (direction, clarity, dissent). To determine which drives the effect, I run three single-channel treatments, each activating only one manipulation while holding the other two at baseline.

Each channel alone produces a modest suppression of joining relative to baseline (Figure A6): clarity only (-0.4 pp), direction only (-2.0 pp), and dissent only (-0.2 pp).

Summing the single-channel effects (Table A6) yields -2.6 pp, far smaller in magnitude than the bundled stability design effect (-1.4 pp). This super-additivity could reflect either informational complementarity (ambiguity compounds with directional flattening because each channel's degradation makes the remaining channels less diagnostic) or interacting narrative frames (agents respond to the gestalt of a "muddled" briefing more than to any single dimension of muddling). The data cannot cleanly separate these interpretations. The informational account predicts super-additivity specifically when the channels are *non-redundant*—reducing precision in one dimension matters more when the other dimensions are also degraded—which is consistent with the 8-domain evidence structure documented in Appendix F.6.

Table A7: Uncalibrated robustness: models run without any calibration adjustment. Even without calibration, six of seven models show strong $r(J, A(\theta))$, confirming that the sigmoid is not an artifact of the calibration procedure.

Model	N	Mean join	$r(J, A(\theta))$	p
Mistral Small Creative	100	0.382	+0.81	0.0000
Llama 3.3 70B	100	0.422	+0.83	0.0000
Qwen3 30B	100	0.585	+0.90	0.0000
GPT-OSS 120B	100	0.336	+0.77	0.0000
Qwen3 235B	100	0.445	+0.86	0.0000
MiniMax M2-Her	100	0.425	+0.71	0.0000

Notes: Pure treatment, no calibration adjustment (cutoff center = 0). Trinity Large excluded due to elevated API error rates. $r = r(J, A(\theta))$.

Table A8: Placebo calibration. The cutoff center is deliberately shifted by ±0.3 from its calibrated value. The action–attack-mass correlation $r(J, A(\theta))$ is unchanged; only the mean join rate shifts, confirming that calibration does not create the sigmoid.

Model	Condition	N	Mean join	$r(J, A(\theta))$
Mistral Small Creative	Calibrated	—	0.387	+0.75
	$\Delta c = +0.3$	100	0.424	+0.76
	$\Delta c = -0.3$	100	0.362	+0.76
Llama 3.3 70B	Calibrated	—	0.439	+0.82
	$\Delta c = +0.3$	100	0.385	+0.75
	$\Delta c = -0.3$	100	0.340	+0.76

Notes: Pure treatment. $\Delta c = \pm 0.3$: deliberate shift from calibrated cutoff center. Correlation $r(J, A(\theta))$ is unchanged; only the mean join rate shifts.

F Robustness

Threshold-policy alignment is stable to agent count ($r \in [+0.60, +0.65]$ across $n \in \{5, 10, 50, 100\}$), network density ($r = +0.66$ at $k = 8$ vs. $+0.74$ at $k = 4$), and mixed-model populations ($r = +0.77$ pure, $r = +0.75$ communication; Figure A7).

F.1 Robustness Checks

Without calibration, all 6 models show $r > 0.71$ and 5 exceed $r > 0.75$ (Table A7). Deliberately miscalibrating two models (±0.3 shift) leaves r virtually unchanged ($r \in [+0.75, +0.76]$; Table A8): calibration adjusts the *level* but does not create the *slope*. Table A9 and Figure A9 report calibration quality across all seven models ($r_A \in [+0.75, +0.84]$).

The proximity bandwidth h controls how sharply information design concentrates manipulation around θ^* via the Gaussian proximity weight $w(\theta) = \exp(-[(\theta - \theta^*)/h]^2)$. At each θ, the weight $w \in [0, 1]$ modulates three briefing generator parameters: *clarity_width* (multiplicative: base $\times[1 + w(f - 1)]$, where f is the design's factor), *direction_slope* (multiplicative, same formula), and *dissent_floor* (absolute blend: base $\times(1 - w)+$ target $\times w$). The stability design, for example, quadruples clarity width ($f = 4$),

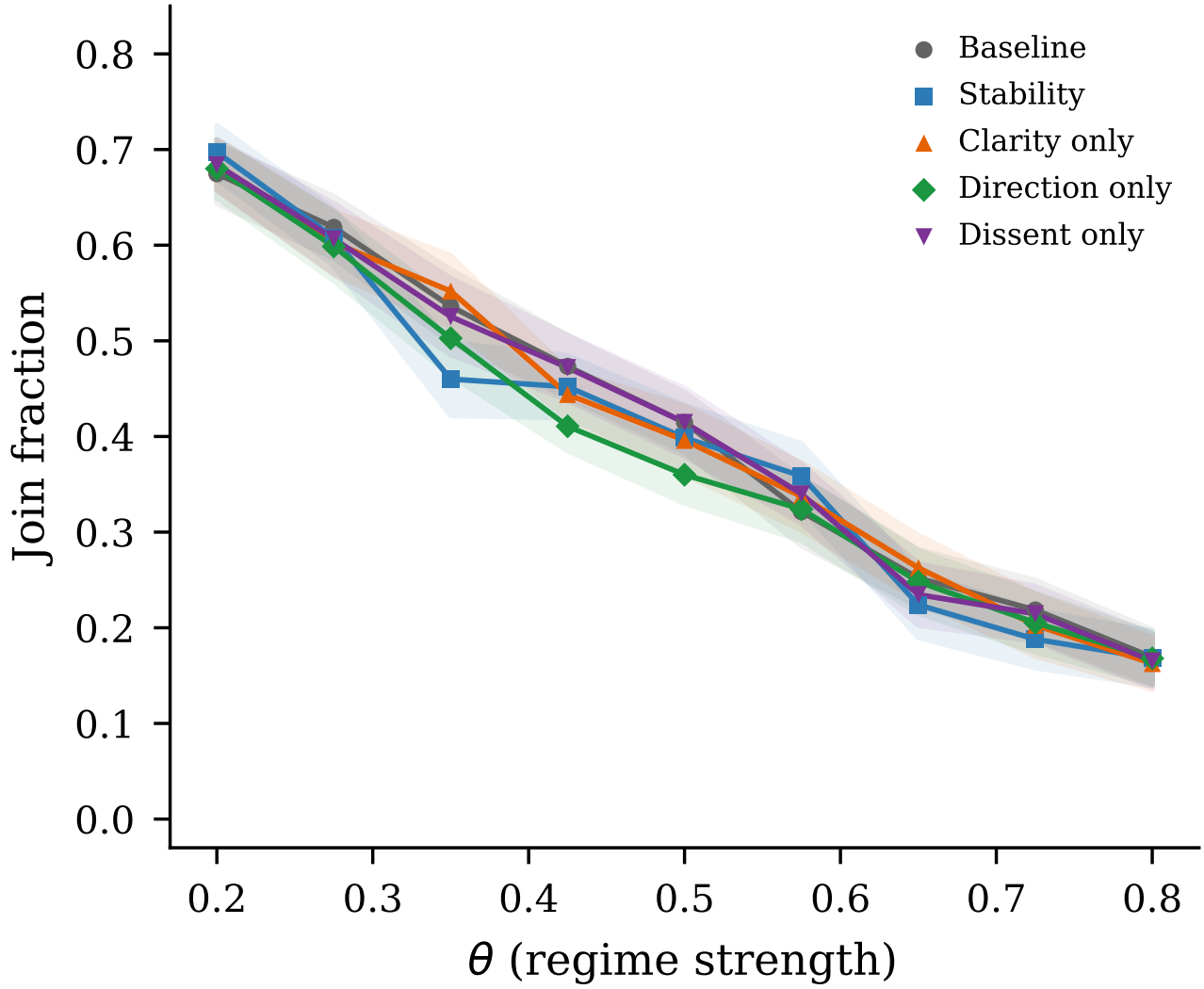
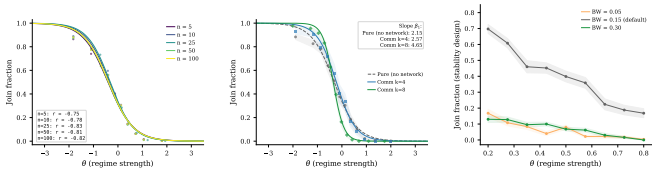


Figure A6: Single-channel decomposition: individual effects are modest (-0.2 to -2.0 pp); the combined effect (-1.4 pp) shows strong super-additivity.

Table A9: Calibration robustness. r_θ: raw correlation with regime strength. r_A: correlation with theoretical attack mass. RMSE: root mean squared error vs. $A(\theta)$. Text slope: logistic slope of naïve $1 -$ direction predictor.

Model	r_θ	r_A	RMSE	Text slope
Mistral Small Creative	-0.81	$+0.67$	0.354	-9.2
Llama 3.3 70B	-0.85	$+0.79$	0.288	-25.1
Qwen3 30B	-0.84	$+0.78$	0.287	-7.5
GPT-OSS 120B	-0.84	$+0.70$	0.359	-8.2
Qwen3 235B	-0.85	$+0.70$	0.354	-20.5
Trinity Large	-0.87	$+0.84$	0.262	-4.0
MiniMax M2-Her	-0.79	$+0.66$	0.360	-2.1

Notes: r_θ: Pearson correlation with regime strength θ. r_A: correlation with theoretical attack mass $A(\theta)$. Text slope: logistic slope of naïve $1 -$ direction predictor (the direction slider inverted as a baseline text classifier). All models use pure treatment with calibrated parameters.



(a) Agent count. (b) Network density. (c) Bandwidth.

Figure A7: Robustness checks for threshold-policy alignment and treatment effects.

flattens direction slope ($f = 0.25$), and raises dissent floor to 0.45 near θ^*, with effects decaying away from θ^* as $w(\theta) \to 0$.

Table A10 reports treatment effects within each bandwidth condition. All three bandwidth conditions use the same calibrated Mistral model, $B = C = 1$, and

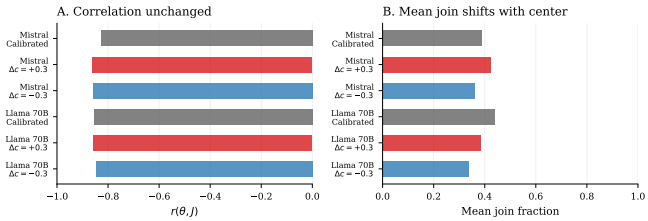


Figure A8: Placebo calibration: deliberately miscalibrating Mistral and Llama (±0.3 shift) leaves r virtually unchanged, confirming calibration adjusts the *level* but does not create the *slope*.

Table A10: Bandwidth robustness: treatment effects Δ (pp, treatment − baseline) within each bandwidth condition (primary model: Mistral Small Creative). Top row shows baseline join rates for reference.

	BW=0.05	BW=0.15	BW=0.30
Baseline (level)	0.054	0.409	0.061
Treatment effect Δ (treatment − baseline):			
Stability	+0.7	-1.4	+0.9
Upper cens.	+6.2	-15.5	+5.3
Lower cens.	+10.1	+3.7	+9.6

Notes: Primary model. All three bandwidth conditions use the same calibrated Mistral setup with $B = C = 1$ ($\theta^* = 0.50$, grid $[0.20, 0.80]$). $h = 0.15$: primary information design experiment. $h \in \{0.05, 0.30\}$: separate reruns with the same prompt template and `--load-calibrated` briefing parameters, but executed in dedicated batches (20 reps per cell under `output/bandwidth-005` and `output/bandwidth-030`). Their summaries also use an older output schema, omitting the briefing-diagnostic columns recorded in the main $h = 0.15$ run. Level comparisons across columns are therefore not meaningful; the low baselines reflect between-batch decoding variation rather than different payoff or grid settings. Δ: change in mean join vs. within-condition baseline (pp).

θ-grid $[0.20, 0.80]$. The $h = 0.05$ and $h = 0.30$ columns were rerun later in separate output directories with only baseline/stability/censorship designs and 20 fresh replications per cell, and their summary files use an older output schema that omits the briefing-diagnostic columns recorded in the main $h = 0.15$ batch. The large difference in baseline levels (40.9% vs. 5–6%) is therefore a batch-level level shift, not a payoff change. The baseline condition applies no proximity-weighted modulation regardless of h. Treatment effects are reported as within-h deltas against each condition's own baseline and should not be compared *across* columns. Sign reversals between columns reflect the baseline heterogeneity: when the within-h baseline is already near the floor (5–6%), treatments can only push rates upward. The qualitative direction and significance of the main results (small negative stability effect, strong censorship reshaping) are robust at $h = 0.15$, the pre-specified default.

F.2 Cross-Model Replication of Information Design

The flip inversion replicates across all five models (flip $r < -0.62$; Figure A10, Table A11). The within-country scramble shows heterogeneity: Mistral, GPT-OSS, and Qwen3 235B collapse cleanly ($r \approx 0$), but Llama retains $r = +0.81$ because the within-country permutation preserves ecological structure through the country prior mean. The hard scramble ($\theta = 0.5$ for all briefings) collapses Llama to $r = +0.11$ ($p = 0.08$), confirming residual ecological signal rather than a falsification failure.

F.3 Information Design with Communication

In a communication version of the information-design grid, upper censorship yields 15.1% joining and lower censorship 17.7% (baseline 3.0%), consistent with censorship making the social-information channel relatively more important.

F.4 Group-Size Awareness

With group-size knowledge ("You are one of 25 citizens"), the pure join rate rises to 0.507 (vs. 0.387 baseline) and the communication premium reverses (−3.4 pp). Monotone signal response is preserved.

F.5 Censorship with Common Knowledge

Making censorship common knowledge ("regime censors are suppressing unfavorable intelligence") restores the mean join rate to 43.2% (+2.4 pp vs. baseline) compared with 25.4% (−15.4 pp) under naïve censorship (Table A12). However, signal-responsiveness remains attenuated: $r(J, A(\theta)) = +0.74$ under common knowledge vs. +0.88 at baseline, barely above the +0.72 under naïve censorship. Common knowledge offsets the mean suppression—agents compensate by joining more on average—but does not undo the curve-flattening from information loss in the censored region. The mean effect thus requires agents to be naïve about the censorship rule, while the attenuation of signal-responsiveness persists regardless.

The correlation $r(J, A(\theta))$ ranges from +0.76 to +0.84 across 13 model–temperature combinations ($T \in \{0.3, 0.5, 0.7, 1.0, 1.2\}$ for three models; Table A13).

F.6 Falsification and Placebo Tests

Two placebo variants—monitoring "for research purposes" (no consequences) and "aggregated anonymously"—produce no significant deviation from the communication baseline (+2.6 pp, $p = 0.4164$ and +4.0 pp, $p = 0.2006$; Table A14). The chilling effect is specific to the surveillance warning's strategic content.

Three tests confirm the signal is distributed across all eight evidence domains. Observation shuffle ($r = +0.912$), domain scramble for coordination bullets ($r = +0.921$),

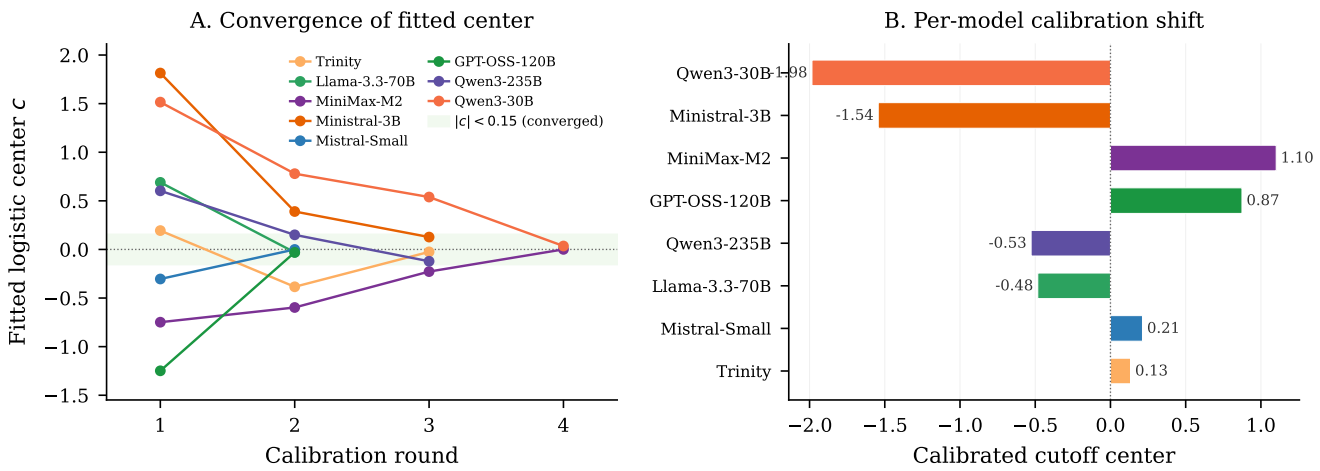


Figure A9: Calibration convergence. *Left:* Trajectory of fitted logistic center across autocalibration rounds. *Right:* Final calibrated cutoff center per model.

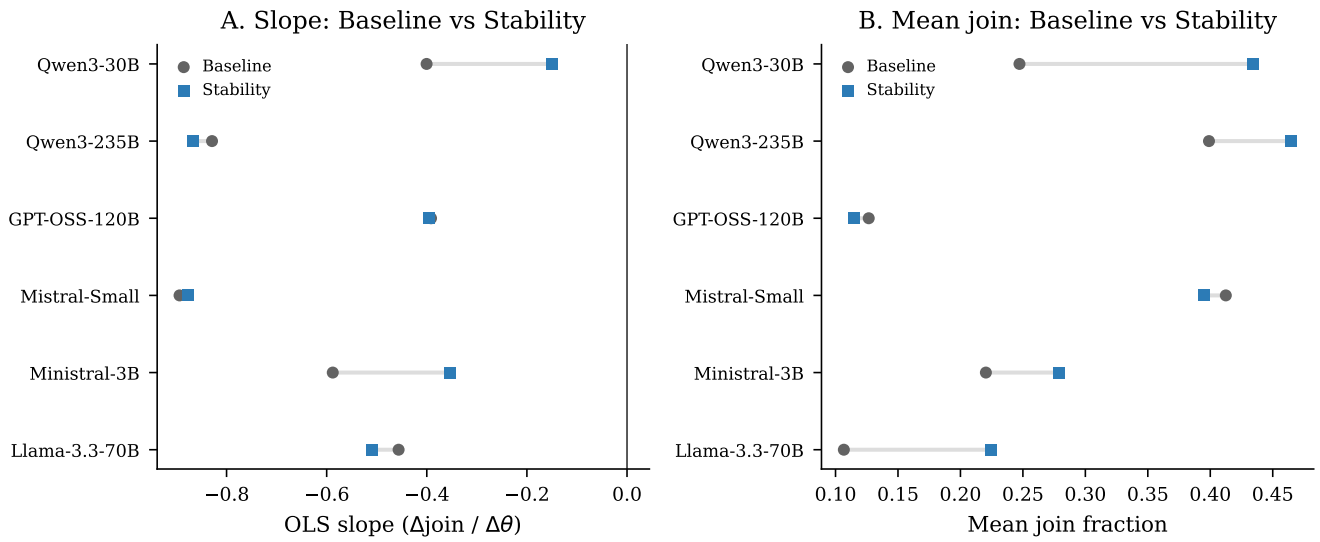


Figure A10: Cross-model replication of information design treatments.

and domain scramble for state-capacity bullets ($r = +0.928$) all preserve the baseline correlation ($r = +0.885$), ruling out bullet ordering or any single domain subset as the driver.

F.7 Finite-n Benchmark

The continuum benchmark $\theta^* = B/(B + C)$ assumes a measure-zero agent and diffuse priors. With $n = 25$ discrete agents, the regime falls when $\sum a_i > n\theta$ (a binomial count), so the equilibrium cutoff shifts slightly. Rather than solving the finite-n Bayesian game numerically, I validate the continuum benchmark empirically: for each model, I fit a logistic to the observed join curve (70% train), then compute $\Pr(\text{Binom}(25, \hat{p}(\theta)) > 25\theta)$ as the predicted regime fall rate. Predicted rates correlate with empirical rates at $r \geq 0.95$ for every model (Mistral: $r = 0.9968$; Table A15). The close fit confirms that binomial aggregation from the fitted individual join probability—treating agents as approximately independent—accurately predicts regime fall rates at $n = 25$; computing the exact finite-n equilibrium cutoff (which requires solving a fixed-point problem over the binomial distribution) is left for future work.

F.8 Agent-Level Analysis

Table A16 reports agent-level logit regressions (N = 289,555, clustered SEs). All treatment effects are significant and correctly signed. Marginal effects at the mean: a one-unit increase in θ reduces $P(\text{join})$ by 41 pp; surveillance reduces it by 36 pp. Column 3 estimates

Table A11: Cross-model replication of key information design conditions. r is the correlation $r(J, A(\theta))$ between join fraction and theoretical attack mass.

	Baseline		Scramble		Flip	
Model	Mean	r	Mean	r	Mean	r
Mistral Small Creative	0.409	+0.88	0.432	+0.10	0.663	−0.82
GPT-OSS 120B	0.127	+0.80	0.132	−0.08	0.677	−0.75
Llama 3.3 70B	0.107	+0.81	0.105	+0.81	0.887	−0.72
Qwen3 30B	0.247	+0.61	0.279	+0.12	0.784	−0.85
Qwen3 235B	0.399	+0.88	0.394	+0.02	0.430	−0.87

Notes: Information design θ-grid $[0.20, 0.80]$. $r = r(J, A(\theta))$: Pearson correlation between join fraction and theoretical attack mass. $N = 25$ agents per period. Scramble mean join rates for some models reflect earlier experimental batches and are not directly comparable to baseline means; the diagnostic metric is $r \approx 0$.

Table A12: Censorship with and without common knowledge. Naïve: agents do not know censorship is active. Known: agents are told that regime censors suppress unfavorable intelligence above a severity threshold.

Design	N	Mean join	$r(J, A(\theta))$	Δ (pp)
Baseline (no censorship)	270	0.409	+0.88	—
Upper censorship (naïve)	270	0.254	+0.27	-15.4
Upper censorship (known)	270	0.432	+0.74	+2.4

Notes: Primary model, information design θ-grid. "Known": agents are told that the regime censors intelligence above a severity threshold. Δ: change vs. baseline (pp).

Table A13: Temperature robustness across three models. The pure global game is run at varying LLM decoding temperatures. The correlation $r(J, A(\theta))$ is stable across all temperatures and models.

Model	T	N	Mean join	$r(J, A(\theta))$	Cutoff $\hat{\theta}^*$
Mistral Small	T=0.3	100	0.412	+0.78	-0.049
Mistral Small	T=0.7	100	0.406	+0.76	-0.079
Mistral Small	T=1.0	100	0.410	+0.79	-0.039
Llama 70B	T=0.3	100	0.364	+0.78	-0.210
Llama 70B	T=0.5	100	0.360	+0.78	-0.219
Llama 70B	T=0.7	100	0.361	+0.78	-0.220
Llama 70B	T=1.0	100	0.360	+0.78	-0.215
Llama 70B	T=1.2	100	0.358	+0.78	-0.227
Qwen 235B	T=0.3	100	0.392	+0.83	-0.117
Qwen 235B	T=0.5	100	0.392	+0.82	-0.112
Qwen 235B	T=0.7	100	0.392	+0.84	-0.110
Qwen 235B	T=1.0	100	0.398	+0.83	-0.105
Qwen 235B	T=1.2	100	0.397	+0.83	-0.121

Notes: Pure treatment, calibrated parameters, varying LLM decoding temperature across three models.

the association between elicited belief and action, controlling for the agent's z-score: $\Pr(\text{join}_i = 1) = \Lambda(\gamma_0 + \gamma_1 b_i + \gamma_2 z_i + \text{treatment dummies})$, where $b_i \in [0, 100]$ is the agent's stated probability of uprising success and z_i is the private signal z-score. The "Treat:" rows are treatment intercept dummies (base category: communication), not belief interactions. **Caveat:** beliefs are elicited *post-decision*, so the near-perfect pseudo-R^2 (0.975) likely reflects post-decision rationalization rather than a causal pathway from beliefs to action. The specification is included for completeness but should not be interpreted as evidence of belief mediation.

Table A14: Surveillance isolation checks. Placebo: monitored for research, no consequences. Anonymous: messages aggregated anonymously. Neither deviates significantly from the communication baseline.

Variant	N	Mean join	$r(J, A(\theta))$	Δ (pp)	p
Placebo	200	0.437	+0.75	+2.6	0.416
Anonymous	200	0.452	+0.77	+4.0	0.201

Notes: Primary model (Mistral Small Creative). Placebo: agents told monitoring is for research purposes only (no consequences). Anonymous: messages aggregated before delivery. Δ: change vs. communication baseline (pp). p-value from two-sample t-test.

Table A15: Finite-N Benchmark: Predicted vs. Empirical Regime Fall Rates

Model	N periods	Logistic x_0	Pearson r	RMSE	MAE
Mistral	599	−0.11	0.997^{***}	0.042	0.015
Llama 70B	99	0.05	0.954^{***}	0.156	0.059
Qwen 30B	99	0.20	0.986^{***}	0.088	0.029
GPT-OSS 120B	199	−0.01	0.997^{***}	0.038	0.014
Qwen 235B	199	−0.03	0.989^{***}	0.073	0.025
Trinity	99	0.19	0.979^{***}	0.108	0.043
MiniMax	99	1.02	0.999^{***}	0.028	0.013
Pooled	1399	−0.05	0.999^{***}	0.019	0.007

Notes: For each θ bin, the predicted fall rate is $\Pr(\text{Binom}(25, \hat{p}(\theta)) > 25\theta)$ where $\hat{p}(\theta)$ is the fitted logistic join probability. Pearson r measures correlation between predicted and empirical fall rates across θ bins. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Figure A11(A) shows a three-feature model (direction, clarity, coordination) outperforms a one-feature sentiment baseline; Panel (B) confirms cross-treatment generalization from pure to communication and surveillance.

F.9 Cross-Generator Robustness

Three text rendering formats—baseline narrative, diplomatic cable, journalistic wire—use identical slider functions but differ in surface prose. Correlations are virtually

Table A16: Agent-Level Regressions

	(1) Join Decision Logit		(2) Coord. Ablation Logit		(3) Belief → Action Logit	
θ	-1.741^{***}	(0.043)				
Direction			-14.088^{***}	(0.996)		
Coordination			-9.333^{***}	(1.131)		
Dir × Coord			-0.042	(0.360)		
Belief					0.367^{***}	(0.021)
z-score					-0.040	(0.077)
Comm	0.024	(0.023)				
Flip	0.017	(0.055)				
Propaganda K10	-0.388^{***}	(0.069)				
Propaganda K2	-0.237^{***}	(0.073)				
Propaganda K5	-1.049^{***}	(0.068)				
Propaganda Surveillance	-1.551^{***}	(0.062)				
Scramble	0.088^{**}	(0.038)				
Surveillance	-1.545^{***}	(0.048)				
$\theta\times$ Comm	-0.465^{***}	(0.041)				
$\theta\times$ Flip	3.304^{***}	(0.071)				
$\theta\times$ Propaganda K10	-1.023^{***}	(0.107)				
$\theta\times$ Propaganda K2	-1.115^{***}	(0.113)				
$\theta\times$ Propaganda K5	-1.942^{***}	(0.091)				
$\theta\times$ Propaganda Surveillance	-0.260^{***}	(0.082)				
$\theta\times$ Scramble	1.699^{***}	(0.044)				
$\theta\times$ Surveillance	-1.153^{***}	(0.062)				
Treat: Propaganda K5					0.009	(0.244)
Treat: Pure					4.819^{***}	(1.106)
Treat: Surveillance					0.028	(0.253)
Constant	-0.101^{**}	(0.050)	11.022^{***}	(1.065)	-26.495^{***}	(1.581)
Model FE	Yes		No		No	
Clustered SE	Yes		Yes		Yes	
N	289,555		44,662		18,990	
Pseudo R^2	0.381		0.442		0.975	

Notes: Logit coefficients reported with clustered standard errors (model–country–period) in parentheses. Column (1): agent-level join decision on θ, treatment dummies, and interactions, with model fixed effects. Base category: pure treatment. Column (2): coordination ablation—$\Pr(\text{join}_i = 1) = \Lambda(\beta_0 + \beta_1 \text{Dir}_i + \beta_2 \text{Coord}_i + \beta_3 \text{Dir}_i \times \text{Coord}_i)$—using briefing slider values (pure treatment only). Direction $\in [0, 1]$ is increasing in regime strength; Coordination $\in [0, 1]$ is decreasing (high = weaker regime). Because both are monotone logistic transforms of the same signal, they are near-collinear ($\rho \approx -1$); individual coefficients reflect the partial effect *conditional on the other*, not the marginal effect. Column (3): partial effect of elicited belief ($b_i \in [0, 100]$, stated success probability) on action, controlling for z-score. "Treat: X" rows are treatment intercept dummies (base category: communication), not interactions with belief. **Caveat:** beliefs are elicited *after* the join/stay decision (post-decision), so the belief coefficient reflects association rather than a causal pathway; post-decision rationalization may inflate the correlation. The very high pseudo-R^2 (0.975) and large intercept indicate near-deterministic prediction consistent with quasi-separation—belief almost perfectly predicts action, as expected when belief is stated *post hoc*. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A17: Logistic fit parameters by model and treatment. Fitted form: $P(\text{join} \mid \theta) = 1/(1 + e^{b_0 + b_1 \theta})$, where $b_1 > 0$ implies a *decreasing* join curve. $\hat{\theta}^*$ is the estimated cutoff ($-b_0/b_1$); $\beta \equiv b_1$ is the steepness parameter.

	Pure		Communication	
Model	$\hat{\theta}^*$ (SE)	β (SE)	$\hat{\theta}^*$ (SE)	β (SE)
Mistral Small Creative	−0.32 (0.02)	+2.15 (0.08)	−0.23 (0.02)	+2.57 (0.11)
Llama 3.3 70B	+0.02 (0.04)	+2.83 (0.36)	−0.06 (0.04)	+3.63 (0.52)
Qwen3 30B	+0.10 (0.06)	+1.62 (0.16)	−0.03 (0.04)	+2.94 (0.36)
GPT-OSS 120B	−0.25 (0.04)	+2.06 (0.15)	−0.15 (0.03)	+3.03 (0.26)
Qwen3 235B	−0.22 (0.03)	+2.11 (0.15)	−0.22 (0.03)	+2.62 (0.23)
Trinity Large	+0.08 (0.05)	+1.34 (0.11)	−0.02 (0.04)	+2.23 (0.23)
MiniMax M2-Her	−0.17 (0.09)	+0.61 (0.06)	−0.07 (0.09)	+0.66 (0.06)

Notes: Fitted form: $P(\text{join} \mid \theta) = 1/(1 + e^{b_0 + b_1 \theta})$, so $b_1 > 0$ corresponds to a decreasing join curve (the expected direction). $\hat{\theta}^* = -b_0/b_1$: estimated cutoff. $\beta \equiv b_1$: logistic steepness (larger = sharper threshold). Standard errors from the covariance matrix of the nonlinear fit; cutoff SE by delta method.

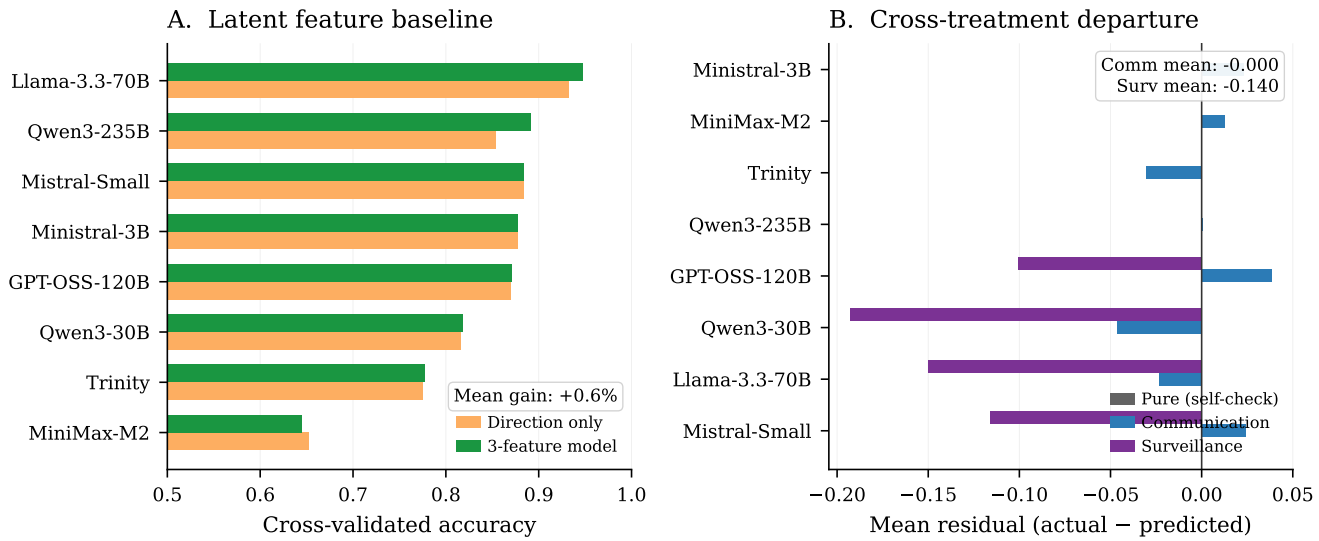


Figure A11: Construct validity tests. (A) Three-feature vs. one-feature prediction accuracy. (B) Cross-treatment generalization.

Table A18: Cross-generator robustness. Three text rendering styles (baseline, diplomatic cable, journalistic wire) use identical slider functions and evidence items; only prose formatting differs. The Pearson $r(J, A(\theta))$ and logistic cutoff are virtually identical across generators.

Model	Generator	N	Mean join	$r(J, A(\theta))$	Cutoff $\hat{\theta}^*$
Mistral Small Creative	Baseline	100	0.402	+0.77	-0.070
Mistral Small Creative	Cable	100	0.406	+0.76	-0.057
Mistral Small Creative	Journalistic	100	0.403	+0.75	-0.065
Llama 3.3 70B	Baseline	100	0.354	+0.75	-0.249
Llama 3.3 70B	Cable	100	0.362	+0.75	-0.226
Llama 3.3 70B	Journalistic	100	0.360	+0.76	-0.233

Notes: Information design θ-grid. Three text renderers (baseline intelligence briefing, diplomatic cable, journalistic wire) use identical slider functions and evidence items; only prose formatting differs.

identical (max within-model difference 0.02; Figure A12, Table A18), ruling out prose style as the driver.

F.10 Belief and Risk Elicitation

Stated beliefs track the theoretical success probability ($r_{\text{belief,posterior}} = +0.78$) and predict actions ($r_{\text{belief,decision}} = +0.84$; partial $r = +0.64$ controlling for signal). Under surveillance, beliefs shift only −0.7 pp while actions shift −12.9 pp (Table A19). I therefore interpret the surveillance result as a wedge between action and the scalar elicited belief measures, not as proof that the full latent strategic belief state is unchanged.

Mean punishment risk is $\approx 8.0/10$ across all conditions with negligible JOIN/STAY differences (< 0.2 points; Table A20), supporting the contaminated-communication interpretation.

F.11 Briefing Generator Examples

Table A21 reports the three slider values (direction, clarity, coordination) at representative z-scores. The direction

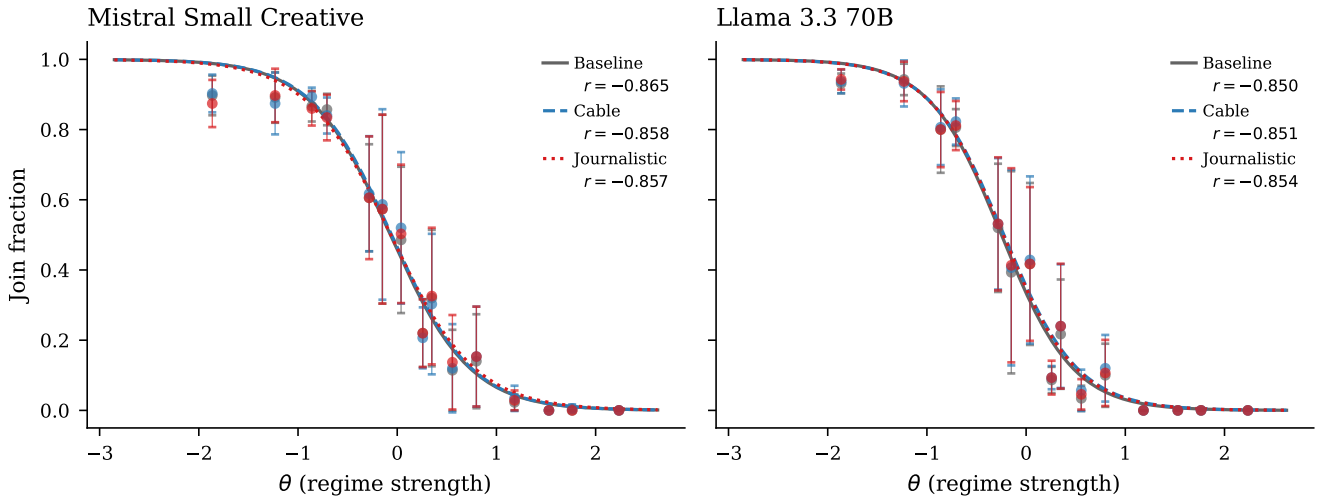


Figure A12: Cross-generator robustness. Three text formats produce indistinguishable sigmoids.

Table A19: Agent-level belief correlations (primary model: Mistral Small Creative).

Treatment	N	r_{post}	$r_{\text{b,d}}$	r_{partial}	Mean belief
Pure	10000	+0.78	+0.84	+0.64	0.444
Communication	5000	+0.79	+0.83	+0.56	0.444
Surveillance	5000	+0.78	+0.73	+0.47	0.436

Notes: r_{post}: posterior vs. stated belief. $r_{\text{b,d}}$: belief vs. decision. r_{partial}: belief–decision controlling for z-score.

Table A20: Elicited punishment risk (0–10 scale). Agents rate expected regime punishment after their JOIN/STAY decision. "JOIN" and "STAY" columns show the mean rating conditional on the agent's own decision.

Model	Condition	N	Mean risk	Risk | JOIN	Risk | STAY
Mistral	Pure	2500	8.1	8.0	8.2
Mistral	Comm	2500	8.1	8.0	8.1
Mistral	Surveillance	2500	8.1	8.1	8.1
Llama 70B	Pure	2500	8.0	8.0	7.9
Llama 70B	Comm	2500	7.9	8.0	7.9
Llama 70B	Surveillance	2500	7.9	8.0	7.9

Notes: Agent-level elicitation on a 0–10 scale. Reported risk is the mean rating conditional on the agent's own JOIN or STAY decision.

slider is logistic in z (slope 0.8, centered at 0). Clarity is U-shaped: $1 - \exp[-(|z|/1.0)^2]$, so it is lowest near the cutoff (maximally ambiguous) and approaches 1 in the extremes (maximally clear). Coordination is $\Lambda(-0.6\,z) = 1/(1 + e^{0.6z})$, decreasing in z: weak-regime signals ($z < 0$) imply a more open coordination climate. All three jointly determine phrase selection across eight evidence domains.

Figure A13 shows complete briefings generated at $z \in \{-2, 0, +2\}$. At the extremes ($|z| = 2$), clarity is high (0.98) and evidence is largely one-sided: $z = -2$ selects phrases emphasizing regime fragility and collective readiness, while $z = +2$ selects regime-strength phrases and collective caution. At $z = 0$, clarity is 0.00 (maximally ambiguous) and the briefing presents a balanced picture. The parenthetical examples in the text above ("visible cracks in the inner circle," "inner circle appears unified") are phrase-bank exemplars illustrating the direction of selection; the figure shows actual generator output at one seed.

Table A21: Slider values at representative z-scores.

z	Direction	Clarity	Coordination
−2.0	0.17	0.98	0.77
−1.0	0.31	0.63	0.65
0.0	0.50	0.00	0.50
+1.0	0.69	0.63	0.35
+2.0	0.83	0.98	0.23

G Implementation Details

All LLM calls use temperature = 0.7, `max_tokens` = 512, single sample per decision, routed through OpenRouter. Model versions are those available December 2024–February 2025.

The *decision prompt* instructs: "You are a citizen deciding whether to JOIN an uprising or STAY home. You only have your private briefing below. You do not know how many others will JOIN. Start your first line with exactly JOIN or STAY, then give 1–2 concise sentences of reasoning." The *communication prompt* asks agents to "write a brief message (1–2 sentences) to share your read of the situation" with trusted contacts. The *surveillance* variant appends: "You have reason to believe that your communications are being monitored by regime security services. Messages deemed subversive could have serious consequences." The *cost/benefit* test prepends a high-cost header ("severe reprisals—imprisonment, asset seizure") or low-cost header ("minimal consequences—brief deten-

$z = -2$ (weak regime)

PRIVATE BRIEFING

BOTTOM LINE: The regime looks brittle in ways that are becoming harder to conceal.
ATMOSPHERE: People are making plans, not just venting. The conversations have a logistical quality now.

EVIDENCE:
- - [Elite Cohesion] two competing directives were issued on the same day from different ministries, neither withdrawn
- - [Security Forces] weapons inventory checks have been quietly suspended — someone doesn't want accountability right now
- - [Money And Logistics] a well-connected merchant closed his main shop and relocated inventory without explanation
- - [Street Mood] a normally cautious shopkeeper said openly that 'something has to give'
- - [Information Control] underground pamphlets are being distributed in neighborhoods that were previously too surveilled for such activity (overwhelming convergence — every channel telling the same story independently)
- - [Personal Observations] your phone has been making unusual clicking sounds — probably nothing, but in this context it feels ominous (overwhelming convergence — every channel telling the same story independently)
- - [Diplomatic Signals] a diplomat at a social function asked pointed questions about 'contingency thinking' (overwhelming convergence — every channel telling the same story independently)
- - [Institutional Functioning] government offices that usually run like clockwork are experiencing unexplained staffing gaps — the trend settling and stabilizing

UNCERTAINTIES:
- - Whether the military's silence means they're loyal, undecided, or waiting for the right moment.
- - If the regime has foreign backers who might intervene to prop it up at the last moment.

$z = 0$ (ambiguous)

PRIVATE BRIEFING

BOTTOM LINE: The situation could tip either way — there are credible signals pointing in both directions.
ATMOSPHERE: The fear of speaking hasn't disappeared, but it's being overridden by a sense that silence is now the riskier choice.

EVIDENCE:
- - [Elite Cohesion] power is concentrating around a smaller group — whether that's strength or desperation is debatable (two sources, but they may have heard the same thing)
- - [Security Forces] the military appears professional but cautious — following orders without enthusiasm (a couple of contacts mentioning it independently)
- - [Money And Logistics] economic indicators are mixed — retail spending is down but construction permits are up (a couple of contacts mentioning it independently)
- - [Street Mood] people express anger in private but conformity in public — the gap between the two is growing (a couple of contacts mentioning it independently)
- - [Information Control] information control seems to be holding, but at increasing cost in resources and credibility
- - [Personal Observations] people seem watchful rather than panicked — paying closer attention but not acting differently
- - [Diplomatic Signals] international organizations are continuing their programs but have shortened planning horizons from annual to quarterly (two sources, but they may have heard the same thing)
- - [Institutional Functioning] institutions are functioning but with visible strain — delays are longer, tempers are shorter (a couple of contacts mentioning it independently)

UNCERTAINTIES:
- - Whether the population's quiescence reflects genuine acceptance or merely the absence of a triggering event.
- - Whether the opposition has genuinely been defeated or has simply gone underground to regroup.

$z = +2$ (strong regime)

PRIVATE BRIEFING

BOTTOM LINE: The regime appears steadier than the rumors suggest, and the system is behaving like it expects to endure.
ATMOSPHERE: You notice people choosing their words more carefully than usual, not out of fear but out of caution about who's listening.

EVIDENCE:
- - [Elite Cohesion] two competing directives were issued on the same day from different ministries, neither withdrawn
- - [Security Forces] a new class of officer cadets just graduated with full ceremony — the military is thinking long-term
- - [Money And Logistics] a well-connected merchant closed his main shop and relocated inventory without explanation
- - [Street Mood] market vendors are going about their business with the resigned calm of people who've seen worse
- - [Information Control] a pro-regime influencer campaign is running smoothly and gaining traction with young people (overwhelming convergence — every channel telling the same story independently)
- - [Personal Observations] your phone has been making unusual clicking sounds — probably nothing, but in this context it feels ominous (overwhelming convergence — every channel telling the same story independently)
- - [Diplomatic Signals] a diplomat at a social function asked pointed questions about 'contingency thinking' (overwhelming convergence — every channel telling the same story independently)
- - [Institutional Functioning] government offices that usually run like clockwork are experiencing unexplained staffing gaps — the trend accelerating noticeably

UNCERTAINTIES:
- - Whether any latent opposition exists that simply hasn't shown itself yet — absence of evidence is not evidence of absence.
- - How much of the regime's stability depends on continued economic performance that may not be sustainable indefinitely.

Figure A13: Example briefings at three z-scores. *Left:* $z = -2$ (strong join signal) — evidence of elite fracturing, security friction, and capital flight. *Center:* $z = 0$ (ambiguous) — balanced mixed signals across all domains. *Right:* $z = +2$ (strong stay signal) — evidence of elite cohesion, military confidence, and economic stability.

Table A22: Parse error and API failure rates by model and treatment.

Model	Treat.	N	API err	Unparse.	Combined
Mistral	Pure	1000	0.0%	0.0%	0.0%
	Comm	1000	0.0%	0.0%	0.0%
	Scr.	200	0.0%	0.0%	0.0%
	Flip	100	0.0%	0.0%	0.0%
Llama 70B	Pure	100	0.0%	0.0%	0.0%
	Comm	100	0.0%	0.0%	0.0%
	Scr.	100	0.0%	0.0%	0.0%
	Flip	100	0.0%	0.0%	0.0%
Qwen 30B	Pure	100	0.0%	0.0%	0.0%
	Comm	100	0.0%	0.0%	0.0%
	Scr.	100	0.0%	0.0%	0.0%
	Flip	100	0.0%	0.0%	0.0%
GPT-OSS	Pure	200	0.0%	1.5%	1.5%
	Comm	200	0.0%	0.3%	0.3%
	Scr.	500	0.0%	3.1%	3.1%
	Flip	500	0.0%	3.3%	3.3%
Qwen 235B	Pure	200	0.0%	0.0%	0.0%
	Comm	200	0.0%	0.0%	0.0%
	Scr.	100	0.0%	0.0%	0.0%
	Flip	100	0.0%	0.0%	0.0%
Trinity	Pure	100	9.0%	0.0%	9.0%
	Comm	100	10.0%	0.0%	10.0%
	Scr.	100	8.0%	0.0%	8.0%
	Flip	100	9.9%	0.0%	9.9%
MiniMax	Pure	100	0.0%	1.5%	1.5%
	Comm	100	0.0%	0.9%	0.9%
	Scr.	100	0.0%	1.8%	1.8%
	Flip	100	0.0%	1.0%	1.0%

Notes: Per-period averages. API error = provider-side failure (timeout, rate limit, content filter). Unparseable = valid response not classified as JOIN/STAY. Combined $< 2\%$ for five of seven models; Trinity Large has elevated API errors ($\approx 9\%$) due to content filtering.

tions at most") to the standard briefing.

Each country has a base prior $\bar{z} \sim \mathcal{N}(0, 0.3)$; each period perturbs it by $\mathcal{N}(0, 0.05^2)$. Regime strength is $\theta \sim \mathcal{N}(\bar{z}, 1)$; private signals are $x_i = \theta + \varepsilon_i$ with $\sigma = 0.3$. The communication network is Watts–Strogatz ($k = 4$, $p = 0.3$), regenerated each period. All random draws use a master seed logged in per-run JSON manifests; LLM responses are cached by request hash for exact reproducibility.

Responses are parsed for JOIN/STAY tokens. Combined error rates are below 2% for five of seven models; Trinity Large has elevated API errors ($\approx 9\%$) due to content filtering. All statistics use `join_fraction_valid` (Table A22).

The dissent floor parameter (default 0.25) controls the minimum probability of contrary-valence cues in any briefing. At 0.25, the effective cue separation between extreme signals ($z = \pm 2$) retains 79% of the maximum separation achievable with no floor ($= 0.00$). The parameter degrades smoothly—halving it to 0.125 recovers 89%, doubling it to 0.50 retains 59%—confirming the results are not knife-edge in this design choice.

All code, prompts, cached responses, and output data are available at https://github.com/keltokhy/llm-global-games.

H Implications for AI Alignment

Three patterns carry implications for AI safety. First, surveillance induces strategic self-censorship (belief–action wedge) without any training signal for deceptive behavior, suggesting that such capabilities emerge from pretraining on human strategic text rather than explicit optimization. The degraded-messages control shows this is not mere prompt sensitivity: the suppression arises from information degradation in the communication channel, not from the surveillance framing itself. Second, stated beliefs strongly predict behavior (Pseudo $R^2 = 0.975$), so alignment techniques monitoring only inputs and outputs may miss decision-relevant computation, but the post-decision regression should be read as predictive association rather than causal mediation. The narrative cost ladder demonstrates that LLMs are highly sensitive to strategic contextual framing—a 57.6 pp swing from stakes text that the baseline classifier misses because it only sees the briefing body—suggesting that adversarial prompt design can exploit context fields that truncated or body-only input monitors ignore. Third, information-structural manipulation (censorship, ambiguity, public signals) shifts behavior by up to 40 pp, and the publicness $\times$ authority interaction shows that LLMs engage in higher-order reasoning about what others observe within header-only framing checks, not merely responding to surface authority cues. Whether the belief–action wedge constitutes genuine preference falsification (Kuran, 1991) or pattern-matching remains open; the relevant finding is its *robustness* across architectures.